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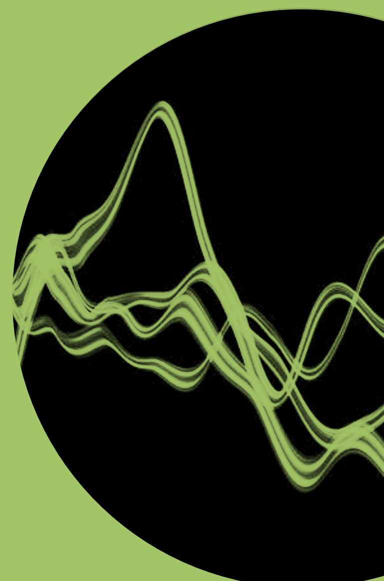
ERICSSON TECHNOLOGY

Review



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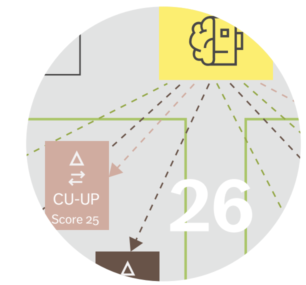

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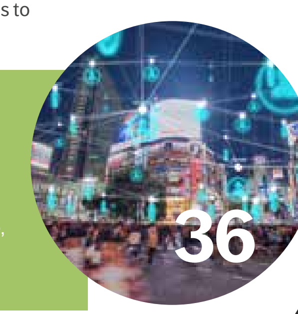
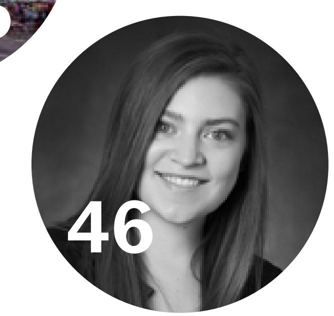


36 FEATURE ARTICLE

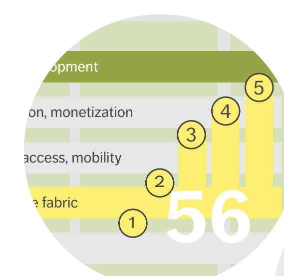
Five network trends towards the 6G era

The cyber-physical convergence is picking up speed, highlighting the need for advanced network technologies to support use cases that blur the boundaries between physical and digital realities. In this year's trends article, CTO Erik Ekudden puts the spotlight on the fundamental network functions related to digital representation.



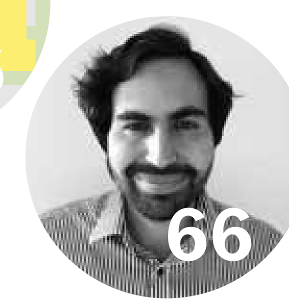
46 ROAMING IN THE 5G SYSTEM: THE 5GS ROAMING ARCHITECTURE

Many network operators around the world are planning to introduce 5GS roaming as a complement to Evolved Packet System (EPS) roaming. Smooth interworking will be essential. Our experts explain how.



56 THE NETWORK COMPUTE FABRIC – ADVANCING DIGITAL TRANSFORMATION WITH EVER-PRESENT SERVICE CONTINUITY

Cloudification and related virtualization of the underlying infrastructure have revolutionized the IT domain, making affordable cloud services available for widespread use. The plans for 6G leverage this development to create the network compute fabric, the innovation platform of the future.



66 XR AND 5G: EXTENDED REALITY AT SCALE WITH TIME-CRITICAL COMMUNICATION

The time-critical communication capabilities in 5G networks will enable major breakthroughs in a wide range of application areas, including extended reality.

Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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Five network trends towards the 6G era
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FULL SPEED AHEAD ON THE JOURNEY TOWARD 6G

■ **THERE IS MUCH** greater awareness today than there was just a few years ago of the essential role that mobile networks play in the digital infrastructure – particularly when it comes to ensuring availability, reliability, sustainability and affordability. In places where 5G networks have become the norm, our vision of mobile networks as the spinal cord of the digital infrastructure is well on its way to becoming reality.

At Ericsson, our long-term goal is to transform the network platform into a powerful innovation platform designed for an intelligent, sustainable and connected world. One important aspect of this work is figuring out how to support use cases that are blurring the boundaries between physical and digital realities. With 5G, we have made significant progress toward enabling physical and digital worlds to converge into an augmented reality that serves communication needs for humans and machines. But expected 6G use cases such as cyber-physical systems and the Internet of Senses will require even more advanced network technologies and capabilities.

With that in mind, this year's technology trends article focuses on the theme of digital representation in future network development. Accurate digital representations of humans, physical objects and surrounding environments are absolutely critical in the convergence of physical and digital worlds. Creating digital representations of sufficient quality to meet the needs of future use cases will require limitless connectivity, trustworthy systems, cognitive networks and what we at Ericsson call the network compute fabric. To learn more about these trends, check out my article on page 36.

> “BLURRING THE BOUNDARIES BETWEEN PHYSICAL AND DIGITAL REALITIES”

This issue of the magazine also includes six other articles highlighting progress in important research areas including extended reality (XR), zero-trust networks, 5GS roaming, data ingestion and energy efficiency. In addition, we have a whole article dedicated to the topic of the network compute fabric, in which the authors explain how we can support distributed intelligence and simplify deployment across heterogeneous infrastructure and administrative domains with the help of unified data access, real-time infrastructure and developer-friendly services.

While all the articles in this issue are worthy of your attention, I want to highlight the XR article in particular. XR technologies have enormous potential to transform both industry and society in the not-so-distant future. Our article explains how mobile network operators can leverage the time-critical communication capabilities in 5G networks to support enterprises that want to launch XR-based applications on a large scale, something that has simply not been possible previously.

We hope you enjoy this issue of our magazine and that you will pass it on to your colleagues and business partners. You can find both PDF and HTML versions of all the articles at: www.ericsson.com/ericsson-technology-review

ERIK EKUDDEN
SENIOR VICE PRESIDENT AND CHIEF TECHNOLOGY OFFICER

Data ingestion architecture for telecom applications

A fast-growing number of business processes and applications rely on data-driven decision-making and telecom artificial intelligence (AI). As new technologies emerge that make it possible to extract additional value from collected data, the opportunities for enhanced analytics and data-driven, AI-powered decision-making continue to expand. To fully capitalize on collected data, communication service providers need data pipelines that are based on a harmonized data ingestion architecture.

ANNA-KARIN RÖNNBERG,
BO ÅSTRÖM,
BULENT GECER

The purpose of data pipelines is to facilitate access to high-quality data for applications ranging from network management and customer experience management (CEM) to business analytics, product serviceability, artificial intelligence (AI)/machine learning (ML) model training, tailored service offerings and AI for Internet of Things solutions. As many of these applications use the same data sets, a harmonized data ingestion architecture boosts efficiency and makes it possible to focus application development resources on use case realizations rather than on data management.

■ Our harmonized data ingestion architecture is built on the idea that data should be collected once and then shared with any authorized application that needs it. The architecture we propose can be deployed both in customer networks and as a service in application clusters (ACs) external to customer networks. It handles data in accordance with customer contracts using automated procedures and provides security mechanisms with controlled access to original data and de-identified data sets.

Data discovery and the ability to control quality and data life cycle are inherent capabilities of the harmonized data ingestion architecture that build trustworthiness for the benefit of every application that consumes the data. Compliance with standards and compatibility with evolving architectures, including ONAP (the Open Network Automation Platform) and ORAN (the Open Radio Access Network), are of crucial importance.

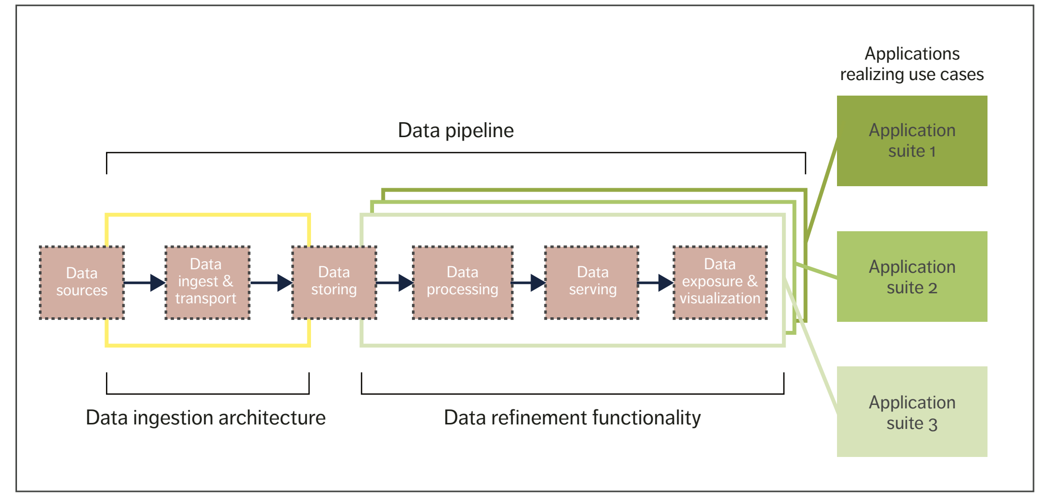

Figure 1 Data pipeline using the harmonized data ingestion architecture

Data pipeline functionality areas

As shown in *Figure 1*, a data pipeline consists of two main functionality areas: data ingestion and data refinement. One conventional approach to data pipelines has been for each application suite to have its own data pipeline, with its own data ingestion and data refinement functionality. To overcome the inefficiencies of this approach, our goal has been to create a harmonized solution for data ingestion in which any collected data is made available for further processing in data-refinement functionality and data-consuming applications. In other words, rather than locking data into one data pipeline or application, we want to make it available to any application suite and its associated data pipeline.

The colored lines in Figure 1 show how the common sets of data collected and made available by the data ingestion functionality (marked in yellow) can be refined in several ways (marked in shades of green) to facilitate and tailor the usage of data in different application suites. At Ericsson, we refer to this approach as "democratizing data."

The benefits of our novel approach to data ingestion include significant opex and capex reductions, increased average revenue per user

Terms and abbreviations

AC – Application Cluster | **AI** – Artificial Intelligence | **ARPU** – Average Revenue Per User | **BDR** – Bulk Data Repository | **CEM** – Customer Experience Management | **DCC** – Data Collection Controller | **DDC** – Data Distribution Central | **DR&D** – Data Routing & Distribution | **DRG** – Data Relay Gateway | **DWH** – Data Warehouse | **EDCA** – Extensible Data Collection Architecture | **GDC** – Global Data Catalog | **ML** – Machine Learning | **ONAP** – Open Network Automation Platform

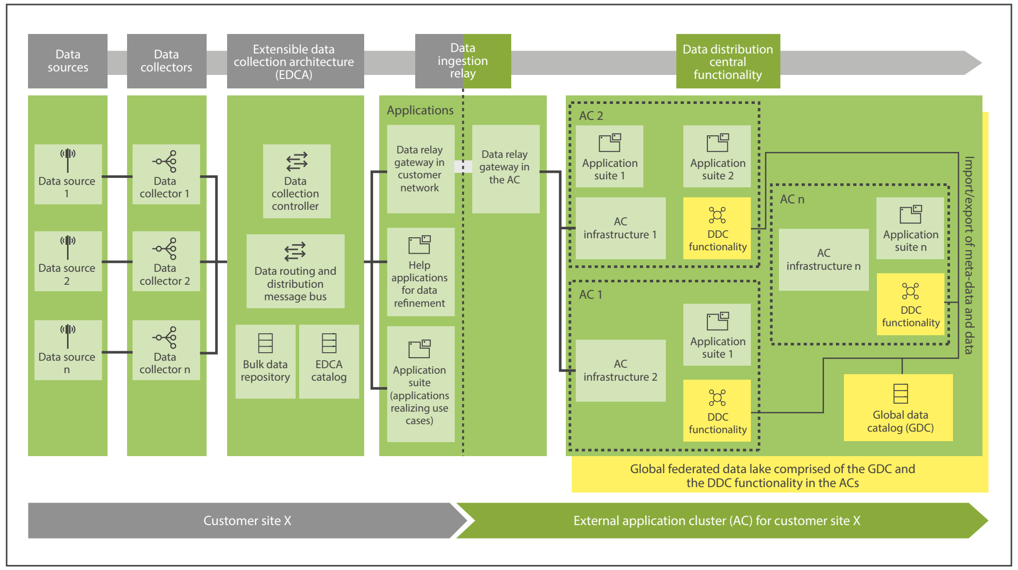

Figure 2 Data ingestion architecture

(ARPU), business process refinement and proactive product support. Opex reductions result from the automation of network assurance, while more efficient network planning and optimization lead to capex savings. The increase in ARPU is due to improvements to network performance and better CEM. Business process refinement comes from increased data quality and faster access to collected data, with the ability to use data collected in near-real time with historical trend analysis. The benefits in terms of proactive product support come from the automation of life-cycle management and serviceability combined with AI-powered decision-making.

Harmonized data ingestion architecture

Put simply, the main objective of our data ingestion architecture is to securely collect data once from the data sources and allow many different consuming applications to use the data in accordance with privacy regulations and customer contracts. These applications may reside in customer networks or be provided as a service and hosted outside the customer networks in application clusters on Ericsson premises or in public clouds, for example.

Figure 2 illustrates the example of a customer network that includes both applications at the customer site (on the left) and applications provided as a service (on the right) as ACs, which are sites where application suites are deployed, typically for a particular purpose such as network operations or product support. Each AC consists of infrastructure and applications where the infrastructure may or may not be cloud based.

Data can be collected from sources for consumption by any application and use case. The collected data is then made available to all applications at the customer site, such as helper applications for data refinement and applications that realize use cases. In accordance with customer contracts, the data ingestion relay transfers data from the customer network to the ACs in Ericsson Cloud using the data relay gateway (DRG). With the help of data distribution central (DDC) functionality and the global data catalog (GDC), the ACs cooperate and form a global federated data lake.

Key functional entities in the data ingestion architecture

The key functional entities in the data ingestion architecture are data sources, data collectors, the extensible data collection architecture (EDCA), the DRG, the DDC functionality, the GDC and the global federated data lake.

Data sources

A data source is a generic term that refers to any type of entity that exports data, such as a node in a telecom network, cloud infrastructure components, a social media site or a weather site. Data sources may export several types of data – including data about performance management, fault management or configuration management – as well as logs and/or statistics from nodes in a telecom network. Data can be exported either by streaming events, files or both.

Data export interfaces will in many cases follow established standards but may also be proprietary. The data can either be pushed from the data source to a data collector or pulled from the data sources by data collectors.

A data source should allow configuration of the data export characteristics, so that it is possible to increase or decrease the data collection frequency, for example. It is not advisable from a load perspective to always collect the maximum number of events from a data source.

From a security point of view, data sources and data collectors should use two-way authentication and encrypt the exported data.

Data collectors

Data collectors facilitate data collection from data sources by using their data export capabilities. They are designed to cause minimal impact to the performance of the data sources. The procedures should be optimized from a data source perspective, and data sets should only be collected once instead of repeatedly. Data can be collected as real-time events or as files in batches depending on the type of data and use cases.

Data collectors make the data available on the EDCA for consumption by multiple applications. It is easy to avoid collecting data that is already available by first checking the EDCA catalog to see which data sets already exist before deploying a new data collector. In this way, data collectors register metadata about their collected data in the EDCA catalog, publish data on data routing and distribution (DR&D) and optionally persist data in the bulk data repository (BDR). Notifications of collected files stored in the BDR are published on DR&D to any consuming application that subscribes to the collected files. The notifications contain references and an address to the BDR that stores the file.

The data collectors collect raw data, and there is an option to apply filtering on the raw data before it is published on DR&D. Further refinement of data can then be tailored for different data pipelines and application suites by EDCA helper applications that implement data-processing and data-serving pipeline functionality, making data ready for visualization and exposure.

Data collectors and EDCA helper applications plug into the EDCA and will evolve according to use case requirements on data pipelines to serve the application suites. It is also possible to integrate data collection from other vendors' data sources with the EDCA.

Extensible data collection architecture

The DR&D function in the EDCA is tasked with receiving data publications from data collectors and delivering these to the consuming applications that subscribe to them. Data on DR&D can be streaming events or notifications of files persisted in the EDCA's BDR. Applications that process the data they receive from DR&D can publish their insights (processed data) back to DR&D, so other applications can benefit from them.

The message bus functionality in DR&D can be realized with one or more message bus technologies, such as Apache Kafka for ONAP compliance, but DR&D is not limited to Kafka. Abstractions are used to shield producers and consumers from the message bus technologies. The DMaaP (Data Movement as a Platform) service wrappers in ONAP are one example of such an abstraction.

> DATA SCHEMAS CAN BE USED TO DYNAMICALLY LOAD MODELS INTO CONSUMING MODEL-DRIVEN APPLICATIONS

DR&D primarily provides the functionality to publish, subscribe to and deliver data, but it can also persist real-time streaming events published on DR&D to enable historical analytics.

DR&D is agnostic to the format of the transferred data. The EDCA catalog stores references to metadata specifications for data sets published on DR&D to facilitate the interpretation and consumption of data by applications. Metadata specifications contain the data schemas (models), and these can be used to dynamically load models into consuming model-driven applications.

The EDCA catalog enables data consumers to discover available data, with instructions on how to access it for consumption. Data must be interpreted to be consumed, and for this purpose the catalog also contains references to metadata specifications for data sets available on the EDCA. Examples of the kind of metadata stored in the catalog include:

- type of data, such as performance management data
- class of data (raw data or insight)
- source of origin
- data quality indication
- data lineage information, including all transformations that have occurred
- location
- data (set) reference
- logical name (URL) for data consumption (DR&D topic)
- schema and serialization method, such as Avro
- metadata specification references, such as references to schemas stored in a schema repository.

The BDR provides the capability to store large amounts of structured, semi-structured and unstructured data – that is, data sometimes referred to as big data. The BDR exposes an (application programming) interface to allow consuming applications that have been notified on DR&D to retrieve a file by using the address and reference included in the notification.

Note that the BDR is not a traditional data warehouse (DWH) intended for long-term data storage. If consuming applications need longer-term storage, subsets of the data should be copied from the BDR and stored in application-specific, long-term DWHs.

The data collection controller (DCC) plays a notable role in the EDCA. Its purpose is to instruct the data sources on how to export data: specifically, it tells them if they should perform basic or extended export. If several applications request extended data collection, it is also the DCC's responsibility to determine when basic data collection can resume and to instruct the data sources to do so. The DCC works in conjunction with data collectors dedicated to specific data sources and implements the configuration of these according to the interfaces and capabilities provided by data sources.

A use case example where the DCC is involved for extended data collection may look something like this:

1. An application interrogates the catalog and discovers that extended data collection from data source 1 is possible but deactivated.
2. The application requires access to the extended set of data and requests that the DCC initiate extended collection through the data collector for data source 1.
3. The DCC supervises the extended reporting.
4. The DCC switches off the extended data collection when no applications require it.

The data relay gateway

The DRG provides functionality to transfer data from a customer network to an external AC in accordance with customer contracts and legislation. It provides a generic solution for any collected data in a customer network and works in harmony with applications in the customer network that require access to the same data.

The data ingestion relay functionality in the DRG functional entity is placed in the customer network and in an external AC. These are named DRG-CN (customer network) and DRG-AC (application cluster). DRG-CN implements data export policies in accordance with customer contracts and regulations, and it transfers data to DRG-AC, which is the entrance to the ACs. DRG-CN is installed per customer and DRG-AC can be installed per AC or be shared by several ACs.

The solution centralizes the data ingestion relay functionality into one scalable functional entity (DRG-CN) that implements data export policies for any data transferred from customer networks to an external AC (DRG-AC). This avoids duplication of this functionality in applications that collect data in customer networks when the same data should be exported to external ACs.

The DRG-CN connects to the EDCA to access both streaming data and files as soon as these are available. In fact, the DRG behaves like any EDCA application in the customer network but with a single purpose: to transfer data to external ACs where it can be used for many purposes, such as training AI/ML models, serviceability, analytics and proactive product support, and so on. Together, the DRG-CN and DRG-AC form a trusted and secure mechanism to exchange data where, for example, one secure VPN tunnel can be used for all data collected by data collectors in customer networks, and for applications in an AC.

The DRG-CN also connects to a database that stores customer contracts according to a defined information model that expresses the written contracts in a formal language that can be used to execute policies in the DRG. The DRG only transfers what it is allowed to transfer according to the contracts, and when it performs a transfer it does so in the manner specified by the relevant contract(s). This means that some data elements must be de-identified with the help of data refinement functionality before they can leave the customer networks. Examples of such data elements include customer identities such as IMSI (the International Mobile Subscriber Identity) and MSISDN (the Mobile Subscriber Integrated Services Digital Network).

Data distribution central functionality

DDC functionality facilitates cooperation between ACs, enabling them to discover and share data, thereby forming a virtual federated data lake. DDC procedures can be grouped into two sets: one with procedures for exchanging metadata, and the other with procedures for transferring data. The first set of procedures is used to discover data, specify requested data and receive notifications of updated data. The second set is used for data transfer and specifies how data sets can be exchanged with details needed for such interactions (with SFTP (SSH File Transfer Protocol) according to schema X, addresses, ports and so on). The DDC procedures enable ACs to automatically discover and consume data that exists in other ACs, as long as the customer contracts allow it.

The global data catalog and federated data lake

The GDC contains all the data in all of the ACs, including metadata regarding orders, customers, products and customer networks. Together with the DDC functionalities of all the ACs, the GDC creates a globally federated data lake. The GDC also serves as the root for discovering data available in the data lake.

Access to all subsets of data will be governed by policies based on contractual rights, with the full set of data visible only to applications and people with the right security classification.

> THE DRG ONLY TRANSFERS WHAT IT IS ALLOWED TO TRANSFER ACCORDING TO THE CONTRACTS

Helper applications for data refinement

Data refinement functionality can be implemented with a set of helper applications that process the raw data available on the EDCA. The purpose of having helper applications is to provide supporting functionality for the applications that realize use cases. The helper applications for data refinement shown in Figure 2 realize data pipeline functionality for data storage, data processing, data serving, data visualization and data exposure.

Other examples of helper applications include:

- security helper applications for de-identification of public data and non-public data
- ETL (extract, transform, load) helper applications for the transformation and preprocessing of raw data into filtered and aggregated data sets
- other repositories to persist structured data for data consumers (applications) with certain needs, such as graph databases for knowledge management.

Helper applications have the ability to offer refined data back to the EDCA for consumers to access it – for example, to applications realizing use cases (CEM, for example) or for exposure in service exposure frameworks. Further, applications that realize use cases and process the refined data to produce insights can also publish those insights in the EDCA for consumption by other applications realizing different use cases. The access rights to the insights and any other data set in the EDCA will be controlled with proper security mechanisms for authentication and policy-based authorization.

Conclusion

Modern intelligent applications depend on the availability of large amounts of high-quality data. In the telecom industry, where many of them rely on the same data sets, gathering the same data for different purposes is a waste of time and resources. In an increasingly data-driven world, it does not make sense to lock data into a single data pipeline or application. On the contrary, collected data should be available for use in the data pipeline of any application suite that needs it. At Ericsson, this is what we mean by democratizing data.

Our research shows that by harmonizing the data ingestion architecture of data pipelines, it is possible to ensure that the necessary data for all applications, regardless of where they are deployed, is only collected once. We can then make the data available to any application that needs it with the help of a data relay gateway. This secure and efficient solution provides the significant benefit of freeing up application development resources to focus on use-case realizations rather than data management.

> “ COLLECTED DATA SHOULD BE AVAILABLE FOR USE IN THE DATA PIPELINE OF ANY APPLICATION SUITE THAT NEEDS IT ”

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Further reading

- **Ericsson, Autonomous networks, available at:** *https://www.ericsson.com/en/future-technologies/autonomous-networks*
- **Ericsson, Future network security, available at:** *https://www.ericsson.com/en/future-technologies/future-network-security*

Ensuring energy-efficient networks

WITH ARTIFICIAL INTELLIGENCE

Finding ways to make networks more energy efficient without negatively impacting QoE is critical to network operators for both cost and sustainability reasons. To assist in these efforts, we are exploring the potential of using artificial intelligence (AI) techniques to recommend energy-efficient configuration settings for network nodes.

KONSTANTINOS VANDIKAS, HELENE HALLBERG, SELIM ICKIN, CECILIA NYSTRÖM, ERIK SANDERS, OLEG GORBATOV, LACKIS ELEFTHERIADIS

Our estimates indicate that the cost of the energy required to power networks represents between 10-30% of the network operating expenses of a communication service provider (CSP), depending on the specificities of its local energy market. In total, this expenditure adds up to approximately 25 billion USD per year [1].

■ Despite the many energy-efficiency solutions already implemented in mobile networks, energy consumption continues to rise in response to the rapid growth of both network traffic and data volumes. Our research indicates that additional energy-efficiency gains can be achieved by using machine-learning (ML) techniques that enable higher levels of automation.

ML is a type of artificial intelligence in which models learn patterns from data without being explicitly programmed. By recommending configuration settings that can be applied to base stations and other equipment, ML-based techniques make it possible to reduce energy consumption in network elements without impacting QoE.

Access nodes are at the center of our work to improve energy efficiency in networks. An access node (often simply called a node) refers to the relationship between connected user devices and the network elements to which those devices are connected. Access node configuration settings strongly influence node energy consumption and potentially many observable network performance QoS metrics.

As configuration settings rarely change at a particular node, the ability to regulate node energy consumption requires a mechanism that enables the generation and evaluation of new configuration settings to explore their impact. To avoid generating configurations that may negatively impact existing key performance indicator (KPI) levels, new configurations need to be bounded by different KPIs that constrain the search.

Given that some configuration settings may require more time than others to take effect, there is a need for accurate predictive models that make it possible to foresee when such changes can be applied ahead of time. This would then minimize any potential disruption to the network's operation. An ideal solution would identify as many potential ways to reduce energy consumption on the current functionality of the network elements.

With all of this in mind, we have developed a concept for end-to-end (E2E) energy optimization that encompasses everything from the power system to the nodes to the network level. *Figure 1* illustrates our concept, highlighting the energy recommendation engine that is at its core.

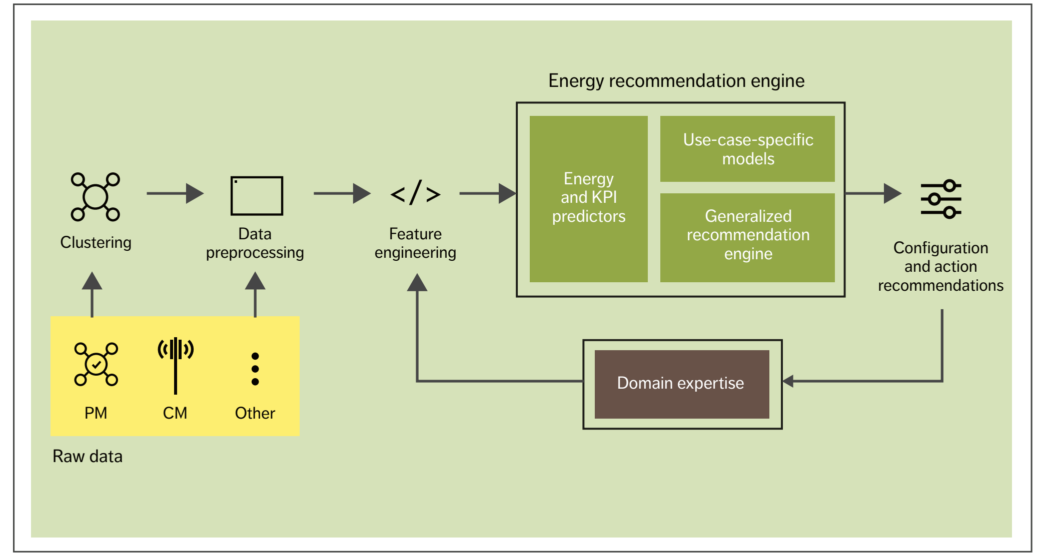

Figure 1 End-to-end energy optimization from power system to node to network

Data set for the energy recommendation engine

All the data used in our work on the energy recommendation engine was collected from a live network. We primarily used performance management (PM) and configuration management (CM) data as measured in the base station, where

Terms and abbreviations

CM – Configuration Management | **CVAE** – Conditional Variational Autoencoder | **DL** – Downlink | **E2E** – End-To-End | **GNN** – Graph Neural Network | **KPI** – Key Performance Indicator | **ML** – Machine Learning | **PM** – Performance Management | **PRB** – Physical Resource Block | **PSU** – Power Supply Unit | **UL** – Uplink

ENERGY-EFFICIENT NETWORKS ✱

TO ENSURE ZERO NEGATIVE IMPACT ON QoE, WE USED KPIs TO CONSTRAIN OUR MODELS

energy measurements are already part of the collected dataset within the PM. As a result, there is no need to deploy new hardware to get the data. The radio network performance counter data sets contain observations on cell performance such as the activity count in downlink (DL) and uplink (UL) directions, the utilization in the cell and units.

To ensure zero negative impact on QoE, we used KPIs to constrain our models. While it is technically possible to include additional and/or alternate KPIs based on operator preferences, we selected these five based on how they affect energy consumption:

1. Number of connection attempts to a cell
2. Average number of users in a cell
3. Throughput
4. Latency
5. Interference.

When telecom networks are installed, they are typically configured with certain parameters such as the number of cells and the hardware unit types (indicating frequency bands) and so on. Possible reasons for a reconfiguration could be a problem such as a software issue or a hardware failure after the installation of new parts that may come from different vendors. Over time, subtle changes and different tuning may lead to different energy consumption levels, where, for the same amount of traffic, this may in some cases be positive (less energy consumption) while in others it is negative (more energy consumption).

The CM data set that we used consists of hundreds of configuration attributes of the sector, including the settings of each radio cell (such as frequency in DL and UL directions) and installed hardware types. We used this information to be able to recommend multiple configuration changes at once, rather than focusing on one at a time.

The output of the energy recommendation engine consists of a set of configuration attribute changes for a corresponding node. The output captures the interplay between different nodes and configurations rather than focusing on isolated fine tuning on a per-node level, which may have an effect on other nodes.

Methodologies used in the energy recommendation engine

A content-based recommendation model requires a good representation of the configuration settings in the embedded (also known as latent) space. Such a representation can be hard to obtain manually due to the high number of configuration attributes.

Conditional variational autoencoders [2] (CVAEs) and graph neural networks [3] (GNNs) are two of the most suitable and complementary techniques for our purposes, as they are both fueled by the success of neural networks. While a CVAE is generative and adversarial, helping to explore large spaces in bounded conditions, a GNN can act as a critic (or discriminator) that can suppress abnormal recommendations, especially when the recommendation model diverges too much.

In addition to GNNs and CVAEs, we also used conditions to confine the new configuration settings under specific KPIs that should not be broken while new configuration settings are created. Such conditions may originate from domain expert engineers, while others can be universal or specific to the network that is being examined.

Graph neural networks

GNNs offer a straightforward way to learn from relational data. We use them – and graph convolution in particular – in two ways in this project: to generate conservative recommendations for energy efficiency based on historical information and to generate multi-site predictive models for different KPIs. Multi-site predictive models are essentially enhanced forecasting models that help operators predict performance based on KPIs that serve as measures of how well a network is behaving.

One of the weaknesses of conventional forecasting techniques such as long-short term memory is the inability to account for the spatio/temporal relationship that exists between nodes. This is problematic because nodes are positioned strategically in various parts of urban or rural spaces and configured accordingly to serve the requests made by user equipment in their vicinity. Information about their spatio/temporal relationships is highly relevant in forecasting.

We have studied the possibility of combining graph convolution with regular 2D convolution. This approach combines two inputs: the adjacency matrix that represents the network's topology, and the time series of each node for a different KPI. The network's topology is constructed using geographical information per node. This information is then represented as an adjacency matrix, which captures the temporal aspects of each node.

Time-series information per node is preprocessed to produce the corresponding input and output predictive window. In this case, we use 12 hours and learn to predict the next 12. Graph convolution and regular 2D convolution is interleaved to build a combination of the two. Our results indicate that this approach increases model performance in multi-site prediction.

The second way that we have used GNNs in this project is to generate conservative recommendations for energy efficiency based on historical information – that is, we used GNNs to create a recommendation engine.

To understand these changes, we represent the relationship between different nodes in a telecommunication network and their configuration sets as a graph. In graph theory, a graph is a structure that contains a set of objects (or nodes) that are connected to each other through links. A link between two nodes means that the two are associated. In this context, we consider a heterogeneous graph, as the objects that we connect are of different types. More specifically, we associate nodes with configuration sets, and the association is represented by a link between them.

In addition, each link is labeled based on the efficiency of that association from an energy perspective as well as on the number of connected subscribers. For each of these elements of the graph we learn a representation of that, driven by its features such as the hardware installed or the different types of parameters in each configuration set.

As a result, the problem is transformed to a link-attribute prediction problem, where we train a model that learns to predict how energy efficient that association was. Even though this system is not capable of generating new configuration sets, we see that it is capable of matchmaking similar nodes with potentially similar configuration sets that can have a lower energy impact than the one currently used.

Conditional variational autoencoder

The high-level definition of a generative model infers generalized distribution of observation features that are confined by the constraint conditions. The CVAE generative model-based recommendation engine consists of multiple components including an encoder, decoder, prediction model for the target KPIs and a prediction model for the energy-consumption target.

The encoder model compresses the representation of the raw CM dataset into a low dimensional matrix, which is referred to as latent space. Latent space consists of points, where each point in a 2D latent space represents a full CM configuration setting that is constrained with an energy consumption and a KPI value.

Due to the nature of the CVAE, the CM configurations with the same energy and KPI constraint categories are located close to each other in the latent space. The decoder reconstructs the complete configuration file, which could consist of many CM attributes (configuration parameters and settings), from the embedding representation in the latent space. The latent variables that represent the targeted KPI and energy-consumption levels are drawn by uniform random selection from the many that represent the same category of the targeted KPI and energy levels. The decoder then uses the input to generate new configuration attributes.

In deployment, we provide constraints as input to the decoder of the CVAE model bounded by the

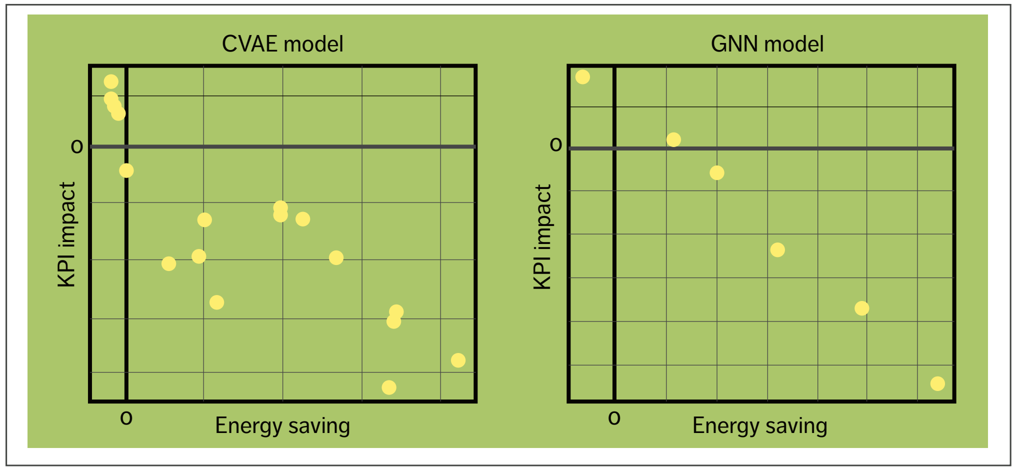

Figure 2 Energy savings and KPI impact according to the CVAE model (left) and the GNN model (right)

required KPI and energy values. We can use multiple constraints and they can be customer specific. The constrained energy value is set slightly less than the predicted energy consumption, as an energy-efficient synthetic CM file is expected as the output of the generative model.

At the same time, we aim to sustain the KPI value. The predicted KPI value at a network node is therefore given as input to the decoder. There are two different prediction models for KPI and energy models:

1. A CM-based prediction model that only uses CM attributes in tabular form as input
2. A PM-based prediction model that only uses PM counters in time-series form as input.

The time-series PM-based prediction models perform 24 hours in advance to give enough time to deploy the desired configuration to a corresponding network element. Prediction results from CM and PM models for energy consumption and KPI (connected users) values performed well. This is important for the accuracy of the generative model output, as the outputs of PM-based models are used as inputs to the CVAE generative model together with the selected latent variables mentioned earlier.

It is important to quantify the amount of energy saved with the generated CM configuration set compared with the existing planned configuration set. For that reason, we used a pretrained CM-based energy model, which only takes CM attributes as input features, and this model predicts energy consumption using purely CM attribute value combinations. This means it can contribute to estimating a mean base energy consumption value given a configuration.

First, the configuration generated by the recommendation model using the actual predicted energy as a constraint is given as input to a pretrained CM-based energy model. Next, the configuration generated by the recommendation model using a value that is lower than the predicted energy as a constraint is given as input to the same CM-based energy model. Finally, the difference between the two energy predictions is computed, and an indicative potential saving is quantified.

We repeat the same steps on a pretrained CM-based KPI model to quantify the KPI impact. This allows us to obtain a KPI versus energy trade-off curve as shown in *Figure 2*. In general, these curves tend to show that higher energy savings yield a higher negative impact on the KPI.

Comparison of the conditional variational autoencoder and graph neural network

A comparison of the CVAE and GNN reveals that, given its generative nature, a CVAE produces new configuration settings. In contrast, a GNN only identifies potential energy savings as marked by a rating function. In this case, the rating function contains six distinct categories, each represented by a dot on the right in Figure 2. (While a GNN is non-generative in the context of our research, the literature indicates that it can also be used in a generative context.)

The comparison also reveals that the efficacy of the CVAE-based recommendation model is dependent on the accuracy of multiple CM- and PM-based KPI and energy prediction models, which means that all the predictive models need to be accurate simultaneously. As a good side effect, this modular structure potentially makes it easy to troubleshoot during the maintenance of the model performance.

Meanwhile, in the case of GNNs, there is only one graph-based model and it is limited by the number of different configurations that are available or have been applied to that network. The use of a single model simplifies its maintenance process.

As both CVAEs and GNNs are predictive recommendation models, it is possible to omit any recommendations that are predicted to impact network performance and QoE in advance. Both models yielded configuration recommendations that are predicted to achieve up to 10% energy savings when applied.

Use-case examples

To further increase the efficacy of our energy recommendation engine, we enhanced its ability to improve energy efficiency by applying recommendations in two specific use cases: radio signal interference detection and PSU load utilization, as shown in *Figure 3*. Both use cases follow the same three steps highlighted in the middle of the figure:

1. Identification (localization) of nodes with the potential for improvement
2. Modeling of the selected nodes to understand their behavior and predict their states
3. Actions to improve energy efficiency (implementation).

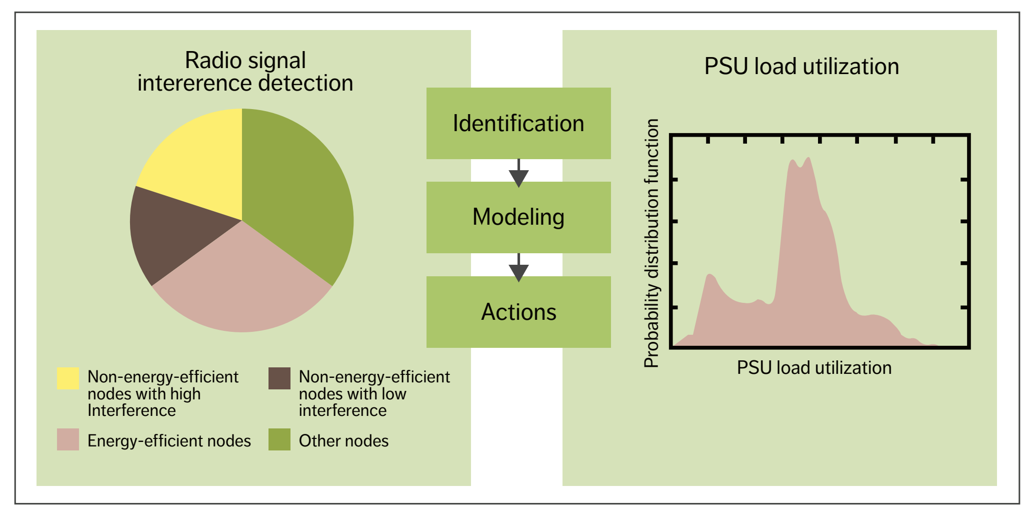

Figure 3 Two use-case examples – radio signal interference detection and PSU load utilization

Use case No. 1: radio signal interference detection

Interference is created by a range of factors such as changes to the environment or having too many users in the same cell, especially if they are located close to the cell edge. It is also possible for cells to interfere in frequency with each other. Interference has a significant impact not only on QoS but also on power consumption. While current radio equipment is designed to handle interference in a way that avoids unwanted emissions and variable techniques are used to limit the interference in the network, radio signal interference has proven to be a formidable challenge to overcome completely.

> A HOLISTIC APPROACH IS THE BEST WAY TO ACHIEVE OVERALL ENERGY SAVINGS

By introducing clusters and dividing the problem into sub-problems, our energy recommendation engine makes it possible to identify the nodes that have high interference and low energy efficiency. The left side of Figure 3 shows a breakdown of radio signal interference into four categories: non-energy-efficient nodes with high interference, non-energy-efficient nodes with low interference, energy-efficient nodes and other nodes. The possibility of improving the nodes classified as other nodes will be addressed in the future.

By modeling the network traffic load, consumed energy and interference-related KPIs, we can save energy by recommending action on the interference cells (such as locking them) to avoid the high energy-consumption state. The cell interference parameters are therefore unique and connected to the network activity and consumed energy.

Use case No. 2: PSU load utilization

A radio base station consists of several PSUs supplying power to the radio units. It is not uncommon for PSUs in radio base stations to be underutilized, as illustrated by the graph on the right in Figure 3. Active but underutilized PSUs may not be working efficiently within the operational range of the unit and consume more power that necessary due to power dissipation. It should be noted that PSU efficiency depends on the load.

To identify nodes for PSU load utilization improvement, we clustered them according to the PSU load, the number of PSUs, relation to the radio network activity (such as the relative radio resource usage), the number of active users, PRB utilization and the number of connected users. We focused on PSUs with less than 50% utilization. Our research show it is possible to propose dynamic power-supply control such as putting underutilized PSUs in sleep mode or turning them off.

PSU efficiency is highly dependent on the load that is applied on the PSU output. Setting one of several PSUs in a system to sleep mode enables savings of 1%. At the same time, the improved utilization and operational efficiency of the PSUs that remain active can provide an additional 1% in savings.

Conclusion

One of our core goals at Ericsson is to continuously improve the energy efficiency of networks. A holistic energy optimization approach is the best way to achieve overall energy savings because it ensures that improvements achieved at one level are not canceled out by increased energy use at another. Our end-to-end (E2E) energy optimization concept is driven by an energy recommendation engine that is powered by artificial intelligence. This solution has great potential for automation, with the help of specific interfaces that can tune the nodes directly without human intervention. It can be fully software based, without the need for additional hardware.

The energy recommendation engine analyzes relevant data to figure out how node configurations can be fine-tuned to reduce energy consumption without impacting QoE. Our research indicates that the total E2E efficiency gains (including radio configurations) generated by our approach can be

up to 10% for radio cells and up to 2% for PSU optimization.

On top of building a generalized energy recommendation engine, we are also developing use-case-specific recommendations for different challenges that can be onboarded to the generalized engine at a later stage. In the two use cases we have studied so far, we used predictive models to find the cases where PSUs are underutilized and to detect interference that may cause unnecessary energy usage.

> USE-CASE-SPECIFIC RECOMMENDATIONS CAN BE ONBOARDED TO THE GENERALIZED ENGINE AT A LATER STAGE

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Further reading

- **Ericsson, AI operations and optimization, available at:** *https://www.ericsson.com/en/ai/operations*
- **Ericsson, AI in networks, available at:** *https://www.ericsson.com/en/ai-and-automation*
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5G zero trust

– A ZERO-TRUST ARCHITECTURE FOR TELECOM

The heterogeneous nature of modern telecommunications infrastructure is making it increasingly difficult to protect network resources with conventional perimeter-oriented approaches to network security. By starting from the assumption that the attacker is already inside the network, the zero trust model enhances security by both blocking unauthorized access to network resources and preventing internal lateral movement by an attacker.

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The primary aim of any approach to network security is to protect the communication infrastructure so that it can provide services with the expected level of quality, free of disruption. By significantly mitigating risks inside the network perimeter, the zero trust model makes it easier for communication service providers (CSPs) to live up to their security commitments.

■ The perimeter security model operates on the basis of inherent trust, assuming that everything on the inside of a network is trustworthy. As long as the attacker is outside the network and the outer perimeter defenses are strong enough to completely prevent breaches, this approach can work well. But if a breach does occur and an attacker gets inside the network, the perimeter security model allows the attacker to move laterally between systems within the network.

The zero trust (ZT) security model resolves this issue by never making any assumptions about trustworthiness. It first emerged more than a decade ago in the enterprise space, which means the telecommunications sector benefits from the enterprise sector's findings and best practices.

A zero trust architecture (ZTA) works by facilitating secure network access to resources (data, devices and services) that is limited only to subjects (users, devices and services) that are authorized and approved. It is built on an identity-centric approach based on the execution of policy-based authorization decisions in runtime combined with traditional defense-in-depth security principles. When implemented correctly, a ZTA mitigates both the risk of an external attacker getting a foothold in the network as well as the risk of lateral movement, in the case of a security breach.

Ericsson's approach to zero trust architecture applies the ZT principles [1, 2] to telecommunications networks. We have chosen to use the terminology and tenets defined by the US National Institute of Standards and Technology (NIST) SP 800-207 [3] (see highlight box). Several other government bodies and organizations are, however, in the process of publishing ZTA guidance or requirements. The National Cyber Security Centre in the United Kingdom currently has its own ZTA design principles [4]. The NSA and US Cybersecurity and Infrastructure Security Agency's Trusted Internet Connections initiative [5, 6, 7] also aligns with ZT principles.

Built-in support for zero trust architecture in 5G

The 3GPP 5G standards define relevant network security features supporting a zero trust approach in the three domains: network access security, network domain security and service-based architecture (SBA) domain security.

Seven tenets for zero trust architecture

The US National Institute of Standards and Technology has defined seven tenets for zero trust architecture [3]:

T1. All data sources and computing services are considered resources. Devices in a network are heterogeneous and they all interact with the network and software services.

T2. All communication is secured regardless of network location. Trust of a device based on where it is located in a network is not enough. All communication should be secure – that is, confidentiality and integrity must be maintained.

T3. Access to individual [operator] resources is granted on a per-session basis. Trust of devices and services is evaluated prior to granting access. Access is ephemeral and only the minimal set of privileges required are granted for the session.

T4. Access to resources is determined by dynamic policy – including the observable state of client identity, application/service, and the requesting asset – and may include other behavioral and environmental attributes. Clients accessing resources are granted access and permissions based on the client's ascertained state and access rules defined in policies.

T5. The [operator] monitors and measures the integrity and security posture of all owned and associated assets. Trust nothing, verify everything. When a request to access a resource appears, the asset is evaluated. The evaluation of assets is continuous, so as to have an accurate assessment of the threat landscape and risks.

T6. All resource authentication and authorization are dynamic and strictly enforced before access is allowed. Authentication and authorization are always required before accessing any resource for a limited time period and this is continual – that is, reauthentication and reauthorization occurs throughout all transactions where required.

T7. The [operator] collects as much information as possible about the current state of assets, network infrastructure and communications and uses it to improve its security posture. Collected data provide context and insights about where security improvements are needed, such as evaluating access requests, optimizing policy creation and enforcement.

THE ABILITY TO SECURELY PROVISION, STORE, ACCESS, USE AND REVOKE CREDENTIALS IMPACTS TRUSTWORTHINESS

The network access security features provide users with secure access to services through the device (mobile phone or connected IoT device) and protect against attacks on the air interface between the device and the radio node.

Network domain security includes features that enable nodes to securely exchange signaling data and user data, for example, between radio and core network functions (NFs) [8].

The 5G SBA is built on web technology and web protocols to enable flexible and scalable deployments using virtualization and container technologies and cloud-based processing platforms. SBA domain security specifies the mechanism for secure communication between NFs within the serving network domain and with other network domains.

Key 5G security features that enable zero trust architecture

In our assessment, there are four key security features in 5G that are of most significance in terms of enabling zero trust architectures: secure digital identities, secure transport, policy frameworks and security monitoring.

Secure digital identities

Identities are the new perimeter to defend in ZT security, as they are the primary factor that determines whether access to resources is granted. Secure digital identities consist of two parts. The first part is the identifier (username, fully qualified domain name, serial number) that uniquely identifies a subject or resource. The second part is the credential (password, private key, token) that is secret data used to verify the authenticity of the subject or resource. The use of secure digital identities must be complemented with processes and technologies that enable the secure management of identities and credentials.

In 5G, each and every subject (subscriber or gNodeB, for example) and resource (such as an SBA NF) is uniquely identifiable. Secure digital identities play a fundamental role in building trust and securing communication between entities across security domains. Examples include the digital identity in SIM cards used to authenticate subscribers and network access control, digital identities based on X.509 certificates used for mutual authentication of network devices and NFs, and management user identities for management access control. Secure digital identities enable the creation of an inventory of network assets and are critical to enabling the authentication of subjects and resources to satisfy NIST tenets T2, T3, T4 and T6.

Confidence in the trustworthiness of an identity is determined by the ability to authenticate the asserted identity and the ability to ascertain the integrity of the device being authenticated. The ability to securely provision, store, access, use, renew and revoke credentials impacts trustworthiness.

Identity life cycle management is more challenging for virtual network functions (VNFs) than it is for network appliances. Firstly, the dynamic nature of virtualized deployments – where NFs are instantiated and removed depending on demand – requires secure provisioning, removal and revocation of digital identities in multi-tenant environments. Secondly, consistent exposure and availability of secure hardware across cloud platforms is needed for secure storage and limited access to key material. Secure hardware is used to protect against theft and misuse of secrets, particularly in multi-tenant environments. Further development and maturity in cloud deployments will be required to protect digital identities and attest the system integrity in 5G networks.

Secure transport

The T2 requirement that all communications must be secured is aligned with 3GPP 5G standards that are developed under the presumption of open networks in which all links could be intercepted. Industry-standard mechanisms are used to secure the communication of user and signaling data across 3GPP interfaces.

Data between the user equipment and the radio base station is secured with cryptographic algorithms, providing confidentiality and integrity protection. Additional improvements are introduced in 5G with the Subscription Concealed Identifier (SUCI) to further enhance protection of subscriber privacy against conventional attacks such as passive eavesdropping or active probing of permanent and temporary identifiers.

Communication in transport networks, and between NFs and interconnect networks, is secured with industry-standard security protocols such as (D)TLS 1.2 and 1.3, IPsec, and MACsec, all of which support mutual authentication.

Policy frameworks

The relationships and interactions between the many logical and physical entities in telecom networks must be managed to ensure that resources are only accessed by authorized subjects. Policies capture the access rules and requirements to determine the eligibility of a request. These policies are managed, distributed and enforced by a policy framework [3, 9]. This enables the enforcement of micro-perimeters with fine-grained access control based on roles, credentials and environmental attributes.

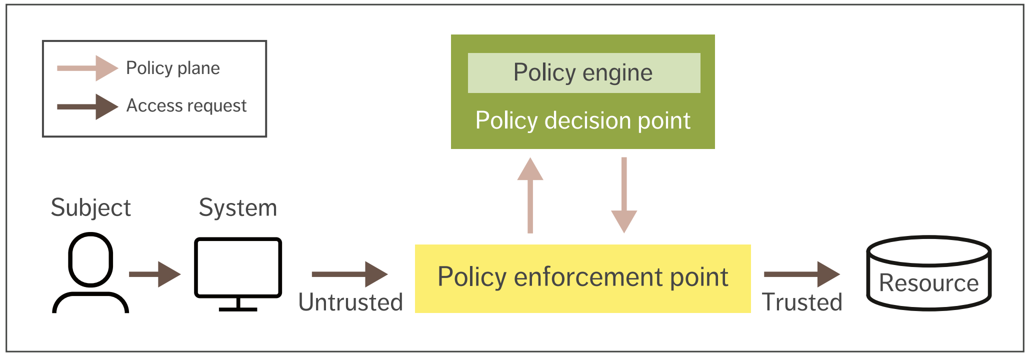

Figure 1 The logical components of the policy framework

Figure 1 presents the logical components of a policy framework. The essential logical entities are the policy decision point (PDP) and policy enforcement point (PEP). To access the specific resource, a subject requests permission from the PDP and provides the information needed to perform authentication and authorization.

Policies are created to reflect an organization's processes and acceptable level of risk as well as the sensitivity of the targeted asset. A policy specifies the required level of protection for an object, privileges of a subject and environmental conditions that can change the allowed behavior of the subject toward the object.

The policy engine is part of the PDP. It runs a trust evaluation algorithm to calculate a subject's trust score, which is used to determine whether the subject is allowed to access a resource. The trust algorithm may only use information provided by the subject or it may utilize additional metadata (geographic location of the subject, historical resource usage and behavior).

The PEP is a component that is responsible for setting up a micro-perimeter to protect a resource. Where possible, the PEP is integrated into the resource or placed as close as possible to it, and it forms a logical demarcation point between security zones. The PEP provides access control of connections between the subject and resource based on access control decisions from the PDP.

Policy frameworks are employed in 3GPP-based systems to manage access to resources in different security domains. For example, to gain access to the 5G network services (T1), the user equipment (UE) contacts an Access and Mobility Management Function (AMF) that takes a PEP role. A PDP role can be represented by multiple NFs where Unified Data Management (UDM) and the Policy Control Function (PCF) may be highlighted, among others.

> THE SECURITY POSTURE OF THE REQUESTING ENTITY MUST BE EVALUATED BY DYNAMIC ACCESS CONTROL POLICIES

The AMF transmits the UE's access request to the UDM to validate the UE's identity and trigger authentication and authorization procedures to establish a secure channel (T2, T6). The PCF feeds the AMF with access and mobility policies that may affect UE authorization to access 5G network resources due to, for example, mobility restrictions (T4) [10, 11, 12, 13].

Another example describes how ZT principles apply in 5G SBA. In reference to T1, the SBA identifies NF service consumers and NF service producers. Communication security between core NFs has improved significantly in comparison with previous generations of mobile networks. SBA security specification requires the performance of Transport Layer Security (TLS) based mutual authentication and OAuth 2.0 token-based authorization for any NF that wants to communicate with another NF (T2, T6). The network repository function (NRF) takes the role of authorization server, which makes the NRF act as the PDP.

The introduction of the service communication proxy (SCP) allows indirect communication between NFs. The SCP can take the role of the PEP and provide access control functionality by requesting authorization decisions from the NRF. This makes it possible to implement the zero trust model in the 5G Core, where an NF service consumer (subject) requests access to an NF service producer (resource) through the SCP (PEP), and the NRF (PDP) grants or denies access [10, 13, 14]. With regard to T4, to support decision-making about requested access to resources, the NRF can store additional information, defining the actions allowed for an NF service consumer to specific NF producers [13].

Security monitoring

Security monitoring supports the detection of threats and measuring the security posture of network assets and compliance with security policies. Monitoring and evaluation of subjects, resources compliance, trustworthiness and state are important when deciding whether to permit access to resources.

The European Telecommunications Standards Institute (ETSI) defines security and trust guidance for NFs [15]. With guidelines emphasizing that compliance and state measurements must be continually monitored to effectively evaluate the level of trust of an NF, ETSI's guidance adheres with the principles of zero trust design.

In line with T3 and T4, the security posture of the requesting entity must be evaluated by dynamic access control policies before access is granted to the requested resource. Additionally, to satisfy T5, all owned assets in a telecom network should be monitored and their security posture should be evaluated continuously. These assets include, but are not limited to, devices accessing the network, RAN NFs, core NFs and management functions.

There are different ways to implement a trust evaluation algorithm. Identifying which trust algorithm implementation to adopt depends on two characteristics:

1. How different parameters are evaluated (as binary decisions or as weighted parts of a whole score or confidence level)
2. How requests are evaluated in relation to other historical requests by the same subject.

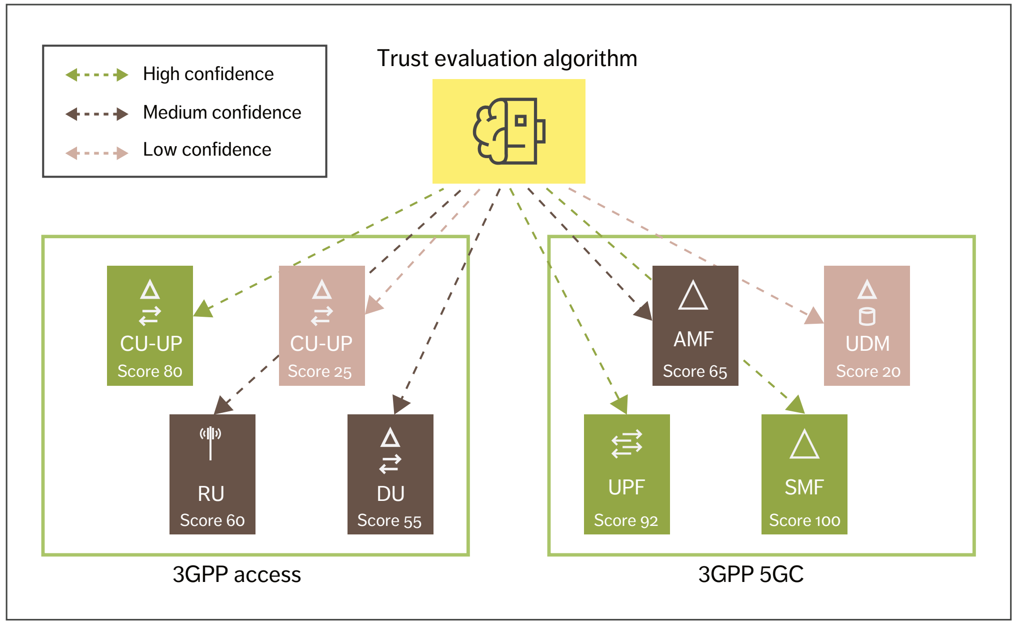

Figure 2 NF trust evaluation

Parameters can be evaluated either based on criteria or score [3]. Score-based evaluation computes a confidence level based on values from every data source, recognizing that there may be various levels of trust between different subjects. Criteria-based evaluation relies on a set of statically configured attributes that must be met before access is granted to a resource or an action is allowed. Moreover, requests can be evaluated either singularly or contextually. Singular evaluation treats each request individually, which risks that an attack can go undetected. Unlike singular evaluations, contextual evaluation takes the subject's history into consideration when evaluating access requests.

The implementation of a trust evaluation algorithm that combines contextual, score-based characteristics would make it possible to offer dynamic and granular access control, since the score provides a confidence level for the requesting account and adapts to changing factors more quickly than static policies.

With respect to T7, there are multiple parameters [15] that can be taken into consideration for evaluating trust that are relevant for telecom networks. Examples include geographical location, NF location, software capabilities (such as patch level, software versions), execution history of an instance, configuration compliance and the appropriate use of encryption techniques.

Future telecom networks should not only consider how to handle trust in subjects, but also trust in resources – particularly in multi-vendor deployments or in cloud where services are provided by a third party.

Figure 2 illustrates how each NF instance may have different trust scores based parameters such as on their current configured state, image version and compliance level. The trust evaluation algorithm in Figure 2 assigns scores in three different ranges on a scale of 0-100: low confidence (<50), medium confidence (51-79) and high confidence (>=80). Based on the evaluation at a certain point in time,

THE TRANSITION TOWARD ZERO TRUST REPRESENTS A MAJOR STEP CHANGE FOR THE TELECOM INDUSTRY

a score is provided to the NF. If the score falls below a certain threshold, further actions should be taken, such as terminating the NF and replacing it.

Next steps

Telecommunications standards have already evolved the telecom security model by adopting ZT principles that better reflect the security reality facing CSPs. While the latest standards provide improved management of security risks for robust and reliable networks and services, a CSP's ability to fully implement a zero trust architecture will also depend on additional technologies, prioritizations and processes.

Consequently the journey towards zero trust will be gradual with methodical decisions on when, where and how to deploy new security technologies and processes. One of the first important decisions a CSP needs to make is whether the transition should include existing infrastructure and, if so, how to include it with minimal operational and security risk. For example, traditional controls should not be decommissioned until careful evaluation and testing of the new security controls has been completed.

The gradual introduction of zero trust principles, process changes and technology solutions should be driven by risk-based decisions about when and where a CSP wants to invest in modernizing its technologies and business processes. Future challenges include the need to manage the risks of both the infrastructure that has migrated to ZTA and the infrastructure that has not.

A successful implementation of ZT builds on the foundation of effective information security and resiliency practices. While a ZTA can help focus security efforts, it is not by itself sufficient to realize a secure architecture. Rather, a ZTA serves as a cornerstone of a holistic active defense strategy for managing risk, complementing established state-of-the-art information security practices.

Today's concept of zero trust, which focuses on network security, will need to evolve in the years ahead. It will need to expand to tackle the issue of how to address vertical trust from the application, the execution environment and device hardware in cloud environments. This includes measuring the system when instantiating network functions and determining the integrity and origin of software.

Additionally, confidential computing technologies to protect software and data will be critical to protect sensitive assets in shared and distributed environments. Hardware rooted security will be essential to establish a verifiable chain of trust from the hardware to the applications that run on it, as well as protecting data in transit, at rest and in use, to address the risks introduced by hardware and software disaggregation and multivendor deployments.

All of these various technical challenges require further research, development and standardization to fully realize the potential of ZTA for the telecom industry.

Conclusion

The transition toward zero trust represents a major step change for the telecom industry. Ericsson is committed to delivering solutions that enable communication service providers (CSPs) to make that transition as smooth as possible. Fortunately, the new requirements and functionality introduced in the 5G specifications are already aligned with many of the zero trust tenets. It is already evident, however, that further technology development, standardization and implementation are needed in areas such as policy frameworks, security monitoring and trust evaluation to support the adoption of zero trust architecture in new telecom environments that are distributed, open, multi-vendor and/or virtualized.

While various technologies can support organizations in adhering to the guiding principles of zero trust as part of their total active defense strategy, it is important to remember that technology alone will never be sufficient to realize the full potential of zero trust. Successful implementation of a network based on zero trust principles requires the concurrent implementation of information security processes, policies and best practices, as well as the presence of knowledgeable security staff. Regardless of where a CSP is in its transition toward a zero trust architecture, the three pillars of people, processes and technology will continue to be the foundation of a robust security architecture.

TECHNOLOGY ALONE WILL NEVER BE SUFFICIENT TO REALIZE THE FULL POTENTIAL OF ZERO TRUST

Terms and abbreviations

AMF – Access and Mobility Management Function | **CSP** – Communication Service Provider | **CU-UP** – Central Unit User Plane | **DU** – Distributed Unit | **ETSI** – European Telecommunications Standards Institute | **NF** – Network Function | **NIST** – National Institute of Standards and Technology | **NRF** – Network Repository Function | **PCF** – Policy Control Function | **PDP** – Policy Decision Point | **PEP** – Policy Enforcement Point | **RU** – Radio Unit | **SBA** – Service-Based Architecture | **SCP** – Service Communication Proxy | **SMF** – Session Management Function | **SUCI** – Subscription Concealed Identifier | **TLS** – Transport Layer Security | **UDM** – Unified Data Management | **UE** – User Equipment | **UPF** – User Plane Function | **VNF** – Virtual Network Function | **ZT** – Zero Trust | **ZTA** – Zero Trust Architecture

Further reading

- **Ericsson, 3GPP 5G security overview, available at:** *https://www.ericsson.com/en/blog/2019/7/3gpp-5g-security-overview*
- **Ericsson white paper, Building trustworthiness into future mobile networks, available at:** *https://www.ericsson.com/en/reports-and-papers/white-papers/building-trustworthiness-into-future-mobile-networks*
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FIVE NETWORK TRENDS

TOWARDS THE 6G ERA

BY: ERIK EKUDDEN, CTO

The pivotal role that the digital infrastructure plays in delivering critical societal, economic and governmental functions has become clearer than ever before as a result of the COVID-19 pandemic. There is now a high level of awareness in both business and society that availability, reliability, affordability and sustainability are all essential aspects of the digital infrastructure that must be ensured in both the short and long term. At the same time, the cyber-physical convergence is picking up speed, highlighting the need for advanced network technologies to support use cases that blur the boundaries between physical and digital realities.

The rapid acceleration in the adoption rate of digitalization during the pandemic would not have been possible without the existing capabilities of both the mobile and the fixed communications infrastructure. Going forward, 5G will be the main digital infrastructure for consumers with mobile and fixed wireless residential access supporting augmented/virtual reality and artificial intelligence (AI) based services. For enterprises – and particularly those that are already moving to cloud services – 5G provides higher performance, edge computing, built-in Internet of Things optimization, workflow automation and network slicing for Service Level Agreement-based QoS. The desire for more advanced use cases in which the physical and digital worlds converge can already be seen in 5G and will be even more evident in the development of 6G. For example, the Internet of Senses (IoS) will augment our senses beyond the boundaries of our bodies. The network will provide cost-effective and trustworthy solutions for these use cases.

Increasing reliance on cloud technologies is going to be essential. Many network operators have already started to include cloud-native technologies in their networks – examples include the use of cloud-native standalone core and cloud RANs with an open layered architecture. Future development in this area depends on the existence of high-performance digital infrastructure functions that ensure an open, interoperable, trustworthy, secure, efficiently automated and safe physical world. Beyond exposing traditional communication services, the network will provide new features such as multisensory digital representations including context awareness and observability to support users with insights and reasoning. The distributed functions of the communication network, such as ubiquitous connectivity embedded with intelligent, real-time compute, will work in synergy with the distributed endpoints and the cloud infrastructures, forming the future capabilities of the digital infrastructure.

The foundational principle for the evolution of the digital infrastructure will be openness in technical and business interfaces, guaranteeing an open marketplace. This openness will be instrumental for independent innovation both within and on top of the digital infrastructure, enabling the future business platform. The ecosystem will evolve through business-driven investments, strong partnerships and openness in the digital infrastructure.

In last year's technology trends article, I stressed the importance of the network as the spinal cord of the digital infrastructure and argued that it is the ideal platform for innovation for an intelligent, sustainable and connected world. This year's article focuses on the fundamental network functions related to digital representation that enable the communication paradigm for the networked reality. I have grouped all of these functions together in trend 1, under the heading digital representation for the networked reality. Trends 2-5 are the main network building blocks that will ensure the provision of the required capabilities and services, namely: adaptable limitless connectivity, integrity of trustworthy systems, federated cognitive networks and a unified network compute fabric.

TREND #1:
DIGITAL REPRESENTATION FOR THE NETWORKED REALITY

With 5G, we already enable physical and digital worlds to converge into an augmented reality that serves communication needs for humans and machines. As humans and physical objects are only able to experience the physical world in a local context, the local presence of sensors, actuators and networks is a crucial enabler. The convergence of physical and digital worlds is enabled by digital representations of both humans and physical objects as well as their environments.

Data from embedded sensors and actuators enables the digital observability of the environment and physical assets. The future network will provide low-level processing of billions of different data streams from sensors and prepare them for applications. The preparation will handle aggregation, filtering and fusion of the data streams. Processes are monitored in real time, and actuators will enable autonomous operations. To further improve observability, the network will generate sensory data such as identity, positioning, time stamps and spatial mapping information. While 5G enables the basic functions, 6G will provide enhancements across these domains. The following functions and capabilities are the most crucial in the development of the future network platform.

Network-aware rendering and synchronization

Some use cases require upper-bound guaranteed end-to-end latency. As the network is context aware, from the capabilities of spatial mapping and dynamic object handling, rendering algorithms provides the dynamic latency and control required to optimize the quality of the user experience. The optimization includes synchronization between all sensory modalities and digital objects in the physical environment.

Collaborative contextual awareness and observability

Connected intelligent machines rely on contextual awareness and observability to relate to, and cooperate with, other machines nearby. This includes the registration and tracking of physical trajectories, intents and capabilities, for example. The network will serve intelligent machines through a context fabric of contextual information including real-time processing of such contextual information. The context fabric thus provides validated information to the collaborative machines about their personas and capabilities, as well as contextual data.

Interoperability among machines and devices

Diversity among connected intelligent machines and devices leads to heterogeneity in terms of semantic descriptions and information formats, and mediation and interoperability are thus required between them. The optimization of communication between machines and devices calls for syntactical and semantic interoperability, as well as programmability between different protocols and models. The network communication stack will provide support for this, thereby off-loading the issue of interoperability.

Real-time positioning

Physical and logical positions as well as time observations will be essential for relating the digital representations to the physical world, thereby generating digital context awareness. Such context-aware data needs to be presented in a uniform and standardized format to enable usage across platforms and to operate and manage systems of systems. This also enables marketplaces for continuously updated observations, models, lessons learned, insights and other digital representations in real time.

A key network feature is the coordination of safe, collaborative autonomous operations by real-time positioning of physical objects in a dynamic environment. The network also provides secure identification and authentication of objects and their related data for integrity and privacy. This enables the secure control and actuation of physical objects, taking into consideration regulations, policies and roles. These network abilities will be key enablers in the foundation of a trustworthy cyber-physical world.

Spatial mapping

Spatial maps are created by collecting and fusing sensor data from all network-connected devices – such as cameras, lidars, radars, gyroscopes, accelerometers, level sensors and pressure sensors. These network-generated maps are used to correctly position the digital objects to the physical environment, thereby enabling multiuser interaction. This network feature will handle relevant spatial mapping processing to reduce form factors, power consumption and cost on IoS devices. Spatial maps will initially focus on visual representation and will, over time, be extended to spatial sound, touch, smell and taste.

Dynamic object handling

Based on spatial maps, the network will perform dynamic object handling and real-time tracking to merge the physical environment and digital content. Dynamic object handling will be particularly important for advanced occlusion, un-occlusion and the continuous update of moving objects in high-resolution spatial maps. For example, in a fire-rescue operation, un-occlusion of smells and temperature could be used to localize a trapped human. Other features include the ability to collect spatial data to generate personalized user experiences and enable context-aware communication.

Embedded data processing and enrichment

A fundamental capability of the future network is a lightweight pervasive data-processing fabric providing the right data pipeline and compute characteristics. As the connected intelligent machines, devices and sensors will generate massive data and information streams, the network will provide adequate processing for the preparation of data, metadata extraction and annotation. Further, in-network stream processing and enrichment such as event detection, filtering, inferencing and learning, as well as sensor fusion, will be provided and exposed. This in-network stream processing will be supported by specialized hardware acceleration embedded in the network. Data formats and new compression algorithms will be optimized for the need of machines rather than for human consumption and processing. The necessary transcoding and compression algorithms will be embedded in the network.

In addition to the network functions and capabilities described above, the realization of digital representation for the networked reality at scale will also require end-to-end solutions across the digital infrastructure of devices, edges, networks and clouds. As digital representations can include sensitive information, standardized, interoperable and secure use of collaborative spatial maps are prerequisites to build trust, making them a good example of an area that will benefit from ecosystem collaboration and innovation. We will promote openness and close collaboration within the global ecosystem to form the future business platform for innovation.

Technological advances in four key areas are essential for the future network to support the convergence of the physical and digital worlds, namely: limitless connectivity, trustworthy systems, cognitive networks and the network compute fabric (trends 2-5).

KEY USE CASES

Three key use cases are driving the networked reality: the digitalized and programmable physical world, the Internet of Senses (IoS) and connected intelligent machines.

1 THE DIGITALIZED AND PROGRAMMABLE PHYSICAL WORLD

In the future, every physical object – including intelligent machines, humans and their environments – will have a digital representation, and the physical world will be fully programmable and automated. The digital representations will manage and process data both individually and collectively for prediction and planning in relation to the physical world. The generated insights will impact the physical world through orchestration, actuation and reprogramming.

2 INTERNET OF SENSES

The IoS makes it possible to blend multisensory digital experiences with local surroundings and interact remotely with people, devices and robots as if they were nearby. Essential components to realize digital sensory experiences close to those experienced in the physical world are visual, audio, haptic, olfactory (smell) and gustatory (taste) sensing and actuation technologies.

3 CONNECTED INTELLIGENT MACHINES

Connected intelligent machines are physical objects and software agents operating and performing tasks in both the digital and physical realms. They are connected to applications, users and each other in collaborative and aggregate structures. As collaborative capabilities advance, the need for communication capacity and functional capabilities increases exponentially, and new digital and diverse interaction patterns will be generated. Further, connected intelligent machines will be increasingly dependent on awareness of both the physical and digital contexts in which they act.

TRENDS 2-5:

BUILDING BLOCKS OF THE NETWORKED REALITY

TREND #2:
ADAPTABLE LIMITLESS CONNECTIVITY

One of the primary goals of 6G access is to deliver adaptable limitless connectivity that ensures the development of an agile, robust and resilient network. Users and applications should focus on the task at hand anytime and anywhere, while the network should be able to adapt and support their needs. Multi-vendor interfaces will ensure openness both in networks and in the ecosystem at large while minimizing system complexity.

Network adaptability

Solutions that ensure dynamic and flexible site deployments are essential for future high-capacity, resilient networks. Different types of nodes, including ad-hoc and non-terrestrial ones, will be seamlessly integrated. For smaller sites with limited reach, the network topology will evolve with multi-hop routing capabilities leading to cost-effective network densification, as dedicated transport links will not be needed.

High-performance, flexible, scalable and reliable transport will allow heterogeneous deployment scenarios for the whole network. For example, it will simplify the integration of distributed and centralized radio access, as well as public and non-public networks. To keep transport networks flexible and manageable, AI-powered programmability will be used for closed-loop automation and multi-service virtualization.

Device and network programmability

Devices will become more future-proof and be able to leverage more advanced network functions through programmability. Device programmability will include downloadable software stacks and configurable AI models. At the same time, the programmable network functionality will support updated and new device features, resulting in optimized customization of individual devices for specific use cases, as well as faster feature development and time to market.

End-to-end availability and resilience

To meet availability and resilience requirements, the network will be simplified, with fewer node-centric deployments and functional separations between the radio access and core. Standardized multi-vendor interfaces will ensure ecosystem openness while minimizing complexity.

Networks and applications will collaborate to ensure end-to-end performance and the provision of the most suitable services to different applications. Resilience mechanisms and end-to-end transport protocols are good examples of functions that support collaboration with applications. These end-to-end support functions will evolve with the needs and requirements of the applications, such as transport protocols that can handle multi-path communication and intelligent congestion control. To support upper-bound latency application requirements, the network will provide predictable latency with low latency variation.

TREND #3:
INTEGRITY OF TRUSTWORTHY SYSTEMS

6G networks will support trillions of embeddable devices and provide trustworthy, always-available connections with end-to-end assurance to mitigate the widened threat space. An essential feature of the network will be to provide secure services by managing and verifying compliance to security, safety, resiliency and privacy demands. Automation that utilizes AI technologies will be used in the whole product life cycle covering development, deployment and operations. AI technologies will also be used for automatic root-cause analysis, threat detection and response against attacks, as well as unintentional disturbances. These technologies will also enhance service availability.

Confidential computing

Secure cloud and edge systems have strict protection requirements for the execution of sensitive code and data in virtual execution environments. Trustworthy confidential computing features will provide isolated execution environments using encryption and cryptographic integrity protection. This will require specialized hardware and remote attestation by users to verify that their applications execute on genuine and properly configured hardware, enabling hardware root-of-trust and secure bootstrapping of the virtual execution environment.

Evolving technologies like homomorphic encryption, which permits users to perform computations directly on encrypted data, as well as multiparty confidential computation, where data privacy between participating parties is secured, will become available.

Confidential computing technologies can also be used to overcome privacy concerns when performing processing on the cloud and edge.

Secure identities and protocols

The network will provide root-of-trust to secure identities in the highly dynamic and distributed virtualized environment. Secure identities form the baseline for identity management and enhancements to privacy-preserving protocol stacks, which will enable the monitoring and compliance verification of the entire network as well as all connected physical objects and software agents.

Zero-trust architecture

A zero-trust architecture is foundational for facilitating secure access to network resources that is limited to authorized and approved clients. Zero trust is an identity-centric approach by which dynamic policy authorization is executed in runtime. Humans supervise the integrity and security posture of all owned and associated assets.

Service availability assurance

Service availability starts with ensuring the integrity and reliability of the hardware and software components and the consistency of behavior of each service in the operating environment with the assigned network resources. One example is the trade-off between system capacity and performance in the radio access to mitigate the effects of environmental and traffic variability for critical services.

A predictive analysis framework for data-driven operations and real-time policy management will provide real-time composite metrics to secure service availability and to inform critical services about functional safety risks. In addition to the network metrics, users can also share their intentions to further enhance their service availability.

PROOF OF CONCEPT

Ericsson is engaged with the automotive industry in a proof of concept for automated transport robots on a wireless factory floor. The solution will move relevant application-related processing from the connected devices into edge compute capabilities. The cloud-based software stack and related services have been significantly adapted for deterministic real-time performance. To fulfill the industrial requirements on availability and reliability, the integration of the solution into the 5G connectivity network is underway. The network compute fabric is embedded with deterministic computation capabilities to manage event-to-actuation in a fault-tolerant manner for safe and accurate operations.

TREND #4:
FEDERATED COGNITIVE NETWORKS

Continuing along the 5G network automation path, the future network will become cognitive by observing and acting autonomously to optimize its performance. Cognitive 6G networks will enable full automation of network management and configuration tasks, allowing operations and maintenance personnel to supervise the network.

A set of distributed intent managers will be embedded in the network. Each intent manager is controlled by intents that specify expectations including requirements, goals and constraints. An intent manager includes cognitive functions that observe the environment under control and draw conclusions from the acquired data. This implies a reasoning process with the purpose of taking action to fulfill the expectations of the intents. Humans will express intents to control the network.

The cognitive functions use AI features to draw conclusions from raw data. Examples of such features include machine-learning models that can produce insights and machine-reasoning capabilities to execute on intents. Further machine-reasoning and multi-agent reinforcement-learning technologies are candidates to enable collaborative closed-loop functionality.

Trustworthy AI ensures that transparency is built into the network, so that humans can understand why certain actions are taken or why certain conditions cannot be met. One technology for implementing trustworthy AI is Explainable AI, an approach that aims to provide an easily understandable explanation of what an AI outcome is along with the rationale behind it. The implementation of cognitive functions is enabled by an underlying data-driven architecture that includes efficient and secure data ingest. An essential part of the intent manager will be the knowledge base, which is where the knowledge of the cognitive functions resides.

TREND #5:
A UNIFIED NETWORK COMPUTE FABRIC

The convergence of internet, telecom, media and information technologies will lead to the creation of a unified, global system of interconnected components. The network compute fabric facilitates unification across ecosystems, application management, execution environments, and exposure of network and compute capabilities. 6G will act as a controller of the network, covering applications ranging from simple devices to advanced cyber-physical systems. At the same time, 6G will integrate storage, compute and communication to create a distributed unified network compute fabric. This fabric will give service providers access to tools and services beyond connectivity that can be made available to the open marketplace.

Future use cases such as the IoS and connected intelligent machines will be highly distributed with demands on guaranteed low deterministic latency, high throughput and high reliability. The network will complement the connectivity offering with an embedded, unified execution environment that hosts and manages these types of use cases. Hardware and software acceleration, time-sensitive communication technologies such as time-sensitive networking and ultra-reliable low latency communication, as well as timing-aware features will enable guaranteed end-to-end timing and reliability.

Considering the billions of sensor and actuator data streams, in-network stream processing will be provided by deterministic compute capabilities available through interoperable application programming interfaces. In conjunction with the preparation of information such as the provenance of data, metadata extraction and annotation, a lightweight pervasive data pipeline and compute fabric will be provided. Further, the network will provide application deployment, scaling and resource allocation in a serverless fashion to support the execution of all the possible sensor and actuator functions of the future.

This compute fabric will evolve around a federated ecosystem involving actors and users of the air interface, the internet, cloud services and devices. The scale of the ecosystem will require wide usage of bilateral agreements between ecosystem actors. The compute fabric will offer a decentralized and broker-less exchange for virtual services, including identity and relations handling. Smart contract technologies such as distributed ledgers will automate contract negotiation between parties supporting as-a-service delivery models like automated sales, delivery and charging.

Current roaming models for voice and data will expand to include smart contracts for the federated ecosystem. Together with our academic partners, we are conducting research to address the security, confidentiality, privacy and provenance of these types of broker-less exchanges.

Conclusion

The digital infrastructure enabled by 5G and 6G has a critically important role to play in enabling the emerging cyber-physical convergence, particularly with respect to its ability to reach a massive scale. Advanced capabilities and features embedded in the network, such as full real-time spatial mapping and context-aware user data, will be foundational to the next wave of digital transformation. The worldwide system of interconnected physical objects and software agents that emerges as a result will make it easy to integrate entities from different domains to form various systems of systems.

The digital infrastructure will enable the continuous optimization of all digitalized solutions for the benefit of society and business, not only to continuously improve efficiency and comfort, but to minimize environmental impacts as well. With all of this in mind, I foresee unprecedented growth and adoption of digitalized solutions in the years ahead, across virtually every sector of business and industry, as well as in the consumer and public domains. The digital infrastructure will grant access to new and increased revenue streams for all enterprises and industries on a global scale.

It is important to note, however, that the innovation required to develop the capabilities to enable intellectual exchanges between humans, machines and software agents demands new forms of collaboration across the ecosystem. Reaching the full potential of this networked reality will therefore require new collaboration frameworks between vendors, industries and society to ensure fairness, trustworthiness and interoperability. 6G offers a broad platform for innovation and is ideally suited to become the information backbone of society.

The networked reality will be based on a global platform with the responsibility to guarantee performance, integrity and openness in technical and business interfaces, ensuring an open marketplace. It will evolve through business-driven investments and collaborations. Ericsson's investments in 5G and 6G represent our primary contribution in the realization of the networked reality. With regard to collaboration, our approach is driven by our firm belief that the development of the future business platform for a converged cyber-physical world requires deep, long-term engagement with the ecosystem. At Ericsson, we are committed to working together with partners old and new to make this happen.

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Further reading

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Roaming in the 5G System:

THE 5GS ROAMING ARCHITECTURE

Many mobile network operators that are deploying the 5G System (5GS) plan to introduce 5GS roaming as a complement to Evolved Packet System (EPS) roaming. Regardless of where they or their roaming partners are on their 5G journeys, every operator will expect roaming to continue to work as well or better than before. Smooth interworking between EPS roaming and 5GS roaming will therefore be essential.

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Roaming extends the coverage of a home operator's services, allowing its mobile users to use those services within another operator's network, which may be in another country (international roaming) or in the same country (national roaming).

■ Voice roaming first emerged in the early days of GSM and was followed by the introduction of data roaming as part of the General Packet Radio System (GPRS). More sophisticated data roaming became available with the Evolved Packet System (EPS), which went on to support voice services as well. All roaming is built on roaming agreements that define where the service is offered, as well as specifying both the technical and commercial components required to enable the service.

While the 5G System (5GS) also supports other use cases such as the Cellular Internet of Things, the industrial Internet of Things and vehicle use cases, we expect mobile broadband (MBB) to be the first 5GS roaming use case. Today, the EPS supports a variety of MBB use cases such as internet access and operator-provided services like voice and messaging. In some cases, EPS deployments have already been upgraded for early support of 5G by non-standalone (NSA) New Radio (NR) in the Evolved Universal Terrestrial Radio Access Network (E-UTRAN) and also support NSA NR when roaming. The 5GS introduces support for NR standalone in next-generation RAN (NG-RAN) and new 5G Core (5GC) [1, 2].

During the migration to 5G when NR coverage is being built out, services requiring wide-area

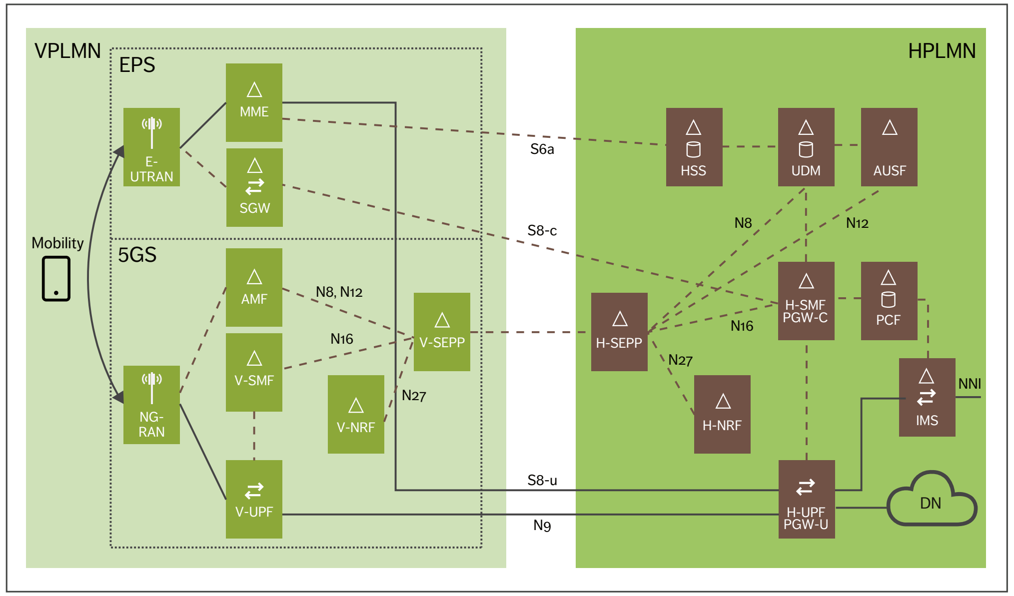

Figure 1 5GS roaming architecture with EPS interworking

coverage are best supported through interworking between the 5GC and the existing Evolved Packet Core (EPC) [3]. In light of this, consideration of the interworking between the 5GC and EPC when roaming is of critical importance.

The GSMA has provided guidelines for both EPS roaming and 5GS roaming [4, 5]. Work on wholesale agreements and solutions related to 5GS roaming is ongoing. The GSMA has also provided guidelines for the user to network interface of Voice over NR (VoNR) [6] and VoLTE [7], including roaming-related functionality.

Home routing (HR) – the main roaming solution for voice and data today – will also be used for 5GS roaming. The 3GPP has also specified a local breakout (LBO) architecture for roaming, but LBO is not used for voice. LBO for data services is rarely used in EPS roaming today and not expected in early 5GS roaming. It should be noted that roaming also requires the user equipment (UE) to support some, if not all, of the frequency bands used by the Visited Public Land Mobile Network (VPLMN) – not only for NR connected to the 5GC but also for LTE connected to the EPC.

Roaming architecture including interworking with the Evolved Packet System

The 5GS roaming architecture enables operators to expand their existing roaming agreements and networks to incorporate 5GS. However, there are both similarities to and differences between EPS roaming and 5GS roaming that operators need to understand. For example, Service-Based Architecture (SBA) and security functionality to protect the network edge are new in the 5GS, supported by a new network function (NF) called the Security Edge Protection Proxy (SEPP).

Figure 1 shows a simplified roaming architecture that includes interworking with the EPS. For the sake of clarity, we have excluded additional NFs such as those for SMS, location handing and so on. While this architecture supports roaming using only the 5GS in the VPLMN and the Home Public Land Mobile Network (HPLMN), it is based on the

> THE ARCHITECTURE… REQUIRES A UE CAPABLE OF USING BOTH THE EPS AND 5GS TO BE ABLE TO ROAM BETWEEN THE TWO

assumption that interworking with the EPS will be required in the initial phases of roaming.

The architecture shown in Figure 1 requires UE capable of using both the EPS and 5GS to be able to roam between the two. The same architecture in the HPLMN can also be used for 4G/NSA UEs and for UEs that are not allowed to use 5GS when roaming, but in those cases only the EPS will be used in the VPLMN.

When the UE connects to the VPLMN, it will register with the Access and Mobility Management Function (AMF). The AMF will query the Network Repository Function (NRF), which in this case serves as a visited-NRF (V-NRF), and the V-NRF will query the home-NRF (H-NRF) to find the Authentication Server Function (AUSF) and the Unified Data Management (UDM) in the HPLMN. Both the traffic between the V-NRF and the H-NRF, as well as all other control plane traffic between the VPLMN and HPLMN, will pass through the SEPPs.

The UE usually sets up one or more protocol data unit (PDU) sessions, which is similar to the non-roaming call flows except that both the V-NRF and H-NRF, as well as the SEPPs, are involved. The usage of a visited Session Management Function (V-SMF) and the relocation of the V-SMF at mobility are specific to roaming – that is, the V-SMF is only used when the UE is in the VPLMN and the PDU session is anchored in the home-SMF (H-SMF) in the HPLMN. In EPS roaming, the EPC nodes serving gateway (SGW) and packet data network gateway (PDN-GW) are used in a PDN connection, regardless of whether the UE is in the VPLMN or the HPLMN.

Terms and abbreviations

5GC – 5G Core | **5GS** – 5G System | **AAA** – Authentication, Authorization and Accounting | **AMF** – Access and Mobility Management Function | **AUSF** – Authentication Server Function | **CDR** – Call Detail Record | **DN** – Data Network | **E-UTRAN** – Evolved Universal Terrestrial Radio Access Network | **EPC** – Evolved Packet Core | **EPS** – Evolved Packet System | **GPRS** – General Packet Radio System | **H-CHF** – Home-Charging Function | **H-NRF** – Home-NRF | **H-PCF** – Home-PCF | **H-SEPP** – Home-SEPP | **H-SMF** – Home-SMF | **H-UPF** – Home-UPF | **HPLMN** – Home PLMN | **HR** – Home Routing | **HSS** – Home Subscriber Server | **IMS** – IP Multimedia Subsystem | **IPUPS** – Inter-PLMN User Plane Security | **IPX** – Internet Protocol Exchange | **LBO** – Local Breakout | **MBB** – Mobile Broadband | **MME** – Mobility Management Entity | **N9HR** – N9 Home Routing | **NAS** – Non-Access Stratum | **NF** – Network Function | **NG-RAN** – Next-Generation RAN | **NNI** – Network-to-Network Interface | **NR** – New Radio | **NRF** – Network Repository Function | **NSA** – Non-standalone | **NSSAA** – Network Slice-Specific Authentication and Authorization | **NSSF** – Network Slice Selection Function | **PCF** – Policy Control Function | **PDN** – Packet Data Network | **PDN-GW** – PDN Gateway | **PDU** – Protocol Data Unit | **PGW** - Packet Data Network Gateway | **PLMN** – Public Land Mobile Network | **PRINS** – Protocol for N32 Interconnect Security | **S8HR** – S8 Home Routing | **S-NSSAI** – Single Network Slice Selection Assistance Information | **SBA** – Service-Based Architecture | **SCP** – Service Communication Proxy | **SEPP** – Security Edge Protection Proxy | **SGW** – Serving Gateway | **SMF** – Session Management Function | **SoR** – Steering of Roaming | **TLS** – Transport Layer Security | **UDM** – Unified Data Management | **UE** – User Equipment | **UPF** – User Plane Function | **V-CHF** – Visited Charging Function | **V-NRF** – Visited NRF | **V-PCF** – Visited PCF | **V-SEPP** – Visited SEPP | **V-SMF** – Visited SMF | **V-UPF** – Visited UPF | **VoNR** – Voice over NR | **VPLMN** – Visited PLMN

Roaming-specific functionality

While the 3GPP first specified 5GS roaming in Release 15, the roaming-related features that it added and/or revised in Release 16 [8] have made Release 16 the baseline for roaming. The most notable roaming-specific functionality in the 5GS falls into six key areas: home control of authentication, roaming restrictions, policy control, charging, QoS control in the V-SMF, and network slicing when roaming.

Home control of authentication

In previous generations, the UE authentication is executed at the serving nodes: the Mobility Management Entity (MME) in the EPC and the mobile switching center/serving GPRS support node in 2G/3G. In the case of roaming, this implies that the execution of UE authentication is delegated to the VPLMN. The delegation of authentication to the MME in a VPLMN is also applicable when a 5G user connects to the EPC while roaming.

The 5GC increases the control of the execution of the UE authentication procedure in the HPLMN, as the UE authentication is always executed and controlled in the AUSF at the HPLMN. Additionally, the AUSF informs the UDM about the result of each UE authentication procedure, so that the UDM can link the result of the authentication with subsequent procedures. This is useful to prevent certain types of fraud, such as fraudulent requests for registering a serving AMF in UDM for subscribers that are not actually present (that is, not authenticated) in the VPLMN.

Roaming restrictions

When a 5GS-capable UE tries to connect over NR to a 5GC when roaming, the VPLMN requests the HPLMN to authorize the inbound roaming UE to connect from that VPLMN before the VPLMN allows the UE to connect to its 5GC. This is referred to as roaming restriction control at the HPLMN as in EPC.

The UDM within the 5GC at the HPLMN determines if the UE is allowed to roam in the VPLMN 5GC. Even if the UE is allowed to roam in the VPLMN 5GC, UE-level roaming restrictions may indicate which HPLMN services can be used while roaming (for example, data services but not voice services). If this is used to restrict the IP Multimedia Subsystem (IMS) voice service when roaming, a voice-centric UE will not connect to the 5GC and will instead look for another radio access in the VPLMN that provides voice service.

Policy control

The home-routed roaming architecture anchors PDU sessions at the H-SMF. As a result, all interactions with the Policy Control Function (PCF) for session management policies take place within the HPLMN domain.

The 3GPP defines the roles of a visited PCF (V-PCF) and a home-PCF (H-PCF) that interact over the N24 reference point for the exchange of UE policies, and of access and mobility policies for roaming UEs.

At least during an initial phase of 5GC roaming, the VPLMN could apply both UE and access and mobility policies for inbound roaming users based on local configuration at the VPLMN without the burden of establishing and maintaining additional roaming reference points for this purpose. This is why the N24 reference point does not appear in Figure 1.

Charging

Both the V-SMF and H-SMF need to support charging. In the VPLMN, the V-SMF interacts with the visited Charging Function (V-CHF), and in the HPLMN the H-SMF interacts with the home-CHF (H-CHF). The visited User-Plane Function (V-UPF) and the home-UPF (H-UPF) need to support the reporting of charging-related session data to the SMF, but the main charging logic resides in the SMFs.

The V-CHF generates CDRs for the inbound roamer traffic, and correspondingly the H-CHF generates call detail records (CDRs) for the outbound roamer traffic.

As a result, the VPLMN has full control over the data volumes that inbound roamers consume in the VPLMN RAN. The VPLMN can also create the necessary data for wholesale (inter-operator) settlement using the same principles as in EPS roaming. The details of the wholesale settlement procedures are described in the GSMA.

QoS control in the visited Session Management Function

In roaming scenarios, any QoS settings requested by the HPLMN should be in accordance with the roaming agreement. However, to protect its network against unwanted resource usage, the VPLMN must have control over and, if necessary, downgrade, the requested QoS [5]. While this is performed by the MME in the EPS, the fact that the 5GS introduces a clear split between mobility management handling (by the AMF) and session management (by the SMF) requires the V-SMF to handle QoS control.

> NETWORK SLICING IS AN INHERENT PART OF THE 5GS THAT IS NOT AVAILABLE IN THE EPS

Network slicing when roaming

Network slicing is an inherent part of the 5GS that is not available in the EPS. When registering in the AMF, the UE determines the network slices it wants to use, expressed as a list of Single Network Slice Selection Assistance Information (S-NSSAI). This list may be empty. The AMF receives the list of subscribed S-NSSAI from the UDM in the HPLMN, and the AMF (possibly assisted by the Network Slice Selection Function (NSSF)) determines which S-NSSAI the UE is allowed to use.

When roaming, it is up to the VPLMN to decide whether inbound roaming UEs use the S-NSSAI provided by the HPLMN or an S-NSSAI provided by the VPLMN. In the latter case, the VPLMN's S-NSSAI is mapped to the one provided by the HPLMN. Both the NSSF and the AMF can determine this mapping. The AMF provides mapping information to the UE during the registration procedure. This is one of the mechanisms that can be used to have a dedicated network slice for inbound roaming UEs from different HPLMNs or to map inbound roaming UEs to a network slice supported by the VPLMN, while fulfilling the service requirements. The mapping can only be performed meaningfully if the VPLMN knows that the service requirements can be fulfilled by the network slice in question.

The UE uses the allowed NSSAI to determine which S-NSSAI to use when establishing a PDU session. In the simplest case, there is only one S-NSSAI in the allowed NSSAI. If so, the UE can include this S-NSSAI when establishing a PDU session, and the AMF uses this S-NSSAI to select the V-SMF and the H-SMF. If the UE does not include an S-NSSAI, the AMF determines the S-NSSAI for this PDU session. This principle is likely enough in initial roaming deployment. If there are more than one S-NSSAI in the allowed NSSAI, the UE needs additional information about which S-NSSAI to use when establishing a PDU session. This additional information can be preconfigured on the UE or can be provided by the HPLMN. For the latter, UE Route Selection Policy has been specified, which can be provided by the H-PCF (through the V-PCF and the AMF) to the UE if needed. In this case, the N24 reference point is required.

In Release 16, the 3GPP has also defined a mechanism for Network Slice-Specific Authentication and Authorization (NSSAA), where connectivity to certain S-NSSAIs is required to be authenticated and authorized by an external authentication, authorization and accounting (AAA) server. If used in roaming, the AMF in the VPLMN triggers NSSAA requests to the external AAA server through an NSSAA Function in the HPLMN.

Steering of Roaming

One of the new features specified for 5GS roaming scenarios is related to the PLMN selection at the UE while roaming. Steering of Roaming (SoR) in a 5GS

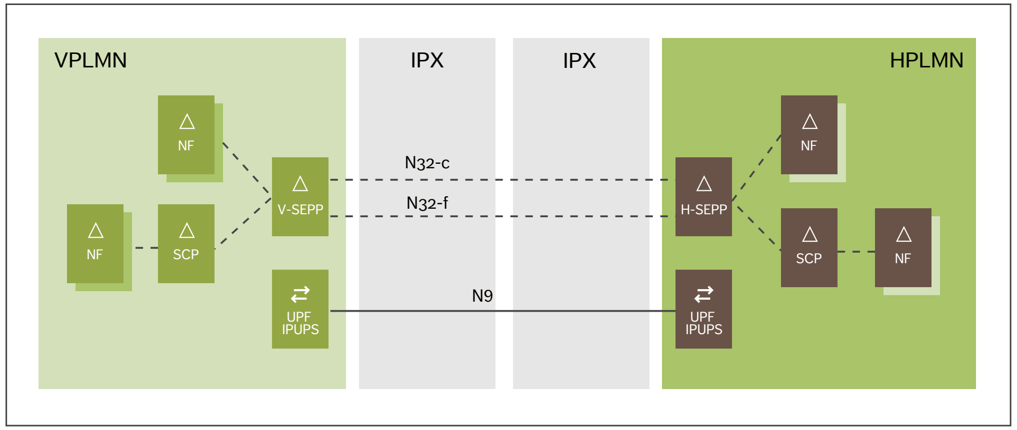

Figure 2 Roaming and SBA

is a control-plane solution that allows the HPLMN to update the UE with the list of preferred PLMN/access-technology combinations. The UE performs PLMN selection based on the received list of preferred PLMN/access-technology combinations. In previous generations, the list of preferred PLMN/access-technology combinations was provided to the UE through Over-the-Air (OTA) mechanisms that were prone to be intercepted and blocked by malicious VPLMNs without the knowledge of the HPLMN. However, not all operators are using SoR today or plan to use SoR in 5GS.

Security in 5GS roaming

5GS includes two proxy types: the Security Edge Protection Proxy (SEPP) for roaming security in SBA and the Service Communication Proxy (SCP) for indirect communication.

Security Edge Protection Proxy

For roaming in SBA, the SEPP secures signaling across PLMN borders by proxying requests and responses for the PLMN, providing topology hiding, signaling firewall, message filtering and additional policy enforcement capabilities. Every control plane message in inter-PLMN signaling passes both the home and the visited PLMN SEPPs. In this way, the SEPP can ensure the protection of messages before sending them to an external network as well as verifying messages received from outside their own network before forwarding them to the appropriate NFs or next-hop SCP. *Figure 2* shows two IPXs (Internet Protocol Exchanges) between the VPLMN and HPLMN, but it is also possible to have just one or even no IPX (in the case of national roaming, for example).

SEPPs authenticate using Transport Layer Security (TLS) over the N32 control plane interface (N32-c), as well as using TLS to protect messages over the N32 forwarding interface (N32-f). Each SEPP must have the credentials of the roaming partner's SEPP. To provide so-called roaming value-added services, the 3GPP has also standardized PRINS (Protocol for N32 Interconnect Security) over N32-f to allow the IPX to add modifications of certain message elements while keeping the original elements. Even if only one side requires its functionality, PRINS requires the support of both the VPLMN and the HPLMN, which means they both have to accept the complexity that PRINS introduces for contracts, operation and security.

While the SEPP provides security for the control plane messages, protection of the user plane messages in inter-PLMN communication is

5GS ROAMING ARCHITECTURE ✱

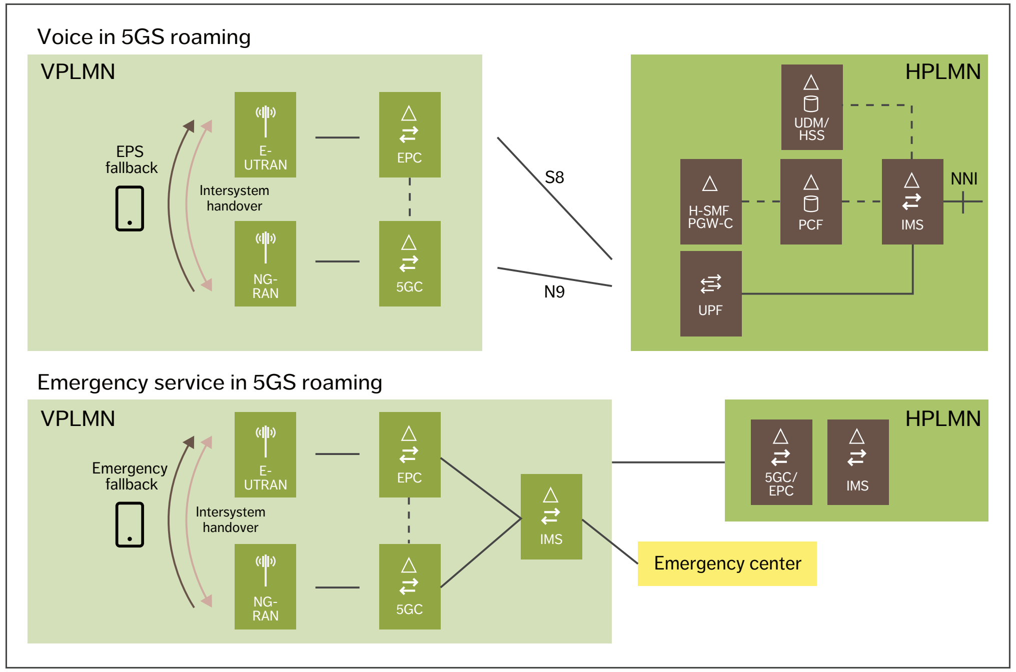

Figure 3 Roaming in 5GS: voice (top) and emergency service (bottom)

provided by the Inter-PLMN User Plane Security (IPUPS) functionality in the existing UPFs, which are controlled by the V-SMF and H-SMF, as shown in Figure 2. IPUPS protects the GTP-U (GPRS Tunneling Protocol-User) traffic by forwarding only valid traffic through the N9 inter-PLMN reference point and discarding the remaining, invalid traffic.

Service Communication Proxy

The SCP was introduced in the 5GS for indirect communication between NFs. SCPs provide centralized monitoring, overload protection and load-balancing functionality. In addition, they provide unified routing and selection logic in determining the destination NF or the next-hop SCP. Routing through the SCP requires support of the 3gpp-Sbi-Target-apiRoot header to indicate the target NF destination. The receiving SEPP may forward a message directly to the destination NF, or through a next-hop SCP, as shown in Figure 2. Likewise, when indirect communication is used, the SCP can support inter-PLMN routing by providing the logic required to route relevant messages to the SEPP centrally. Each operator can decide to deploy SCPs or not, independent of the decision to support roaming.

Voice and other services when roaming

Due to the use of the S8 reference point for routing user traffic and related signaling between the VPLMN and HPLMN, the EPS roaming solution for IMS voice is also known as S8 home routing (S8HR). 5GS roaming for voice follows the same HR principles as for S8HR roaming. 5GS roaming for voice is referred to as N9 home routing (N9HR) and is needed for smartphones. Both EPS fallback and VoNR are possible options as voice solutions.

The IMS PDU session is always anchored in H-SMF/UPF and IMS network elements are in the home network (see Figure 2). Interworking with 4G will be enabled for EPS fallback and for intersystem handover between VoNR and VoLTE. It is recommended to use VoNR when roaming because the additional call setup delay in EPS fallback will be even greater in roaming cases compared with non-roaming ones. Supporting both EPS fallback and VoNR in the HPLMN maximizes the opportunities for compatible network capabilities with roaming partners.

SMS solutions for 5GS roaming are either SMS over IP or SMS over 5G non-access stratum (NAS), and both rely on support for SMS in the HPLMN. SMS over IP has no additional impacts on the VPLMN and is based on the established IMS PDU session to the HPLMN. SMS over 5G NAS depends on the SMSF (the SMS function in 5GC) being present in the VPLMN.

Emergency calls for a roaming subscriber are provided by the VPLMN subject to regulatory requirements. This is applicable to 5GS roaming as well as to S8HR roaming. Emergency service in the 5GS is either provided natively over the 5GS or by emergency service fallback to the EPS. The AMF provides the UE with an indication about which of these two solutions is valid during the 5GC registration procedure. *Figure 3* illustrates the difference between architecture for voice in 5GS roaming (top half) and emergency service in 5GS roaming (bottom half).

For emergency calls on the 5GS, the UE will establish an emergency PDU session. After resolution by the SMF/UPF in the VPLMN, the emergency call will be connected through the VPLMN IMS to the emergency center. In the case of emergency service fallback, mobility from 5GS to the EPS will take place, and then the emergency call will be set up on the EPS using an emergency PDN, with the VPLMN IMS. As in the EPS, since there is typically no IMS network-to-network interface (NNI) between the VPLMN and the HPLMN, the VPLMN IMS cannot fetch UE/subscriber identifiers from the HPLMN IMS. Instead, it must fetch these identifiers from the VPLMN 5GC before emergency call setup. Any emergency callback from the emergency center to the user will be through the HPLMN IMS just as any other incoming call.

> THE INTRODUCTION OF 5GS ROAMING IS GOING TO DEMAND ATTENTION ACROSS ALL DOMAINS

For MBB and data access, the PDU session to the internet (which uses the data network name) is also anchored in HPLMN SMF and UPF. The HPLMN UPF and N6 reference point are used to provide subscribers with internet access (the data network), as shown in Figure 1.

Conclusion

The majority of mobile network operators will create combined Evolved Packet System (EPS) and 5G System (5GS) networks supporting seamless voice and data services in the years ahead. They will continue to expect roaming to work regardless of whether their partners have both the EPS and the 5GS, or only the EPS. While 5GS roaming has many similarities to previous generations of roaming, it also introduces several new concepts that lead to some important differences that all operators need to understand.

One of the main benefits of the 5GS roaming architecture is the possibility to expand an existing EPS roaming solution by using 5GS in the VPLMN and mobility between the 5GS and the EPS when roaming. User equipment that is capable of using both the EPS and the 5GS will also be able to use both EPS and 5GS roaming.

The introduction of 5GS roaming is going to demand attention across all domains. There are roaming aspects to consider in the core network, in user data and policies, in services and in backend systems. At the same time, security for roaming partners must also be ensured. Fortunately, all of these aspects are addressed in the 3GPP Release 16, the new baseline for roaming.

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Further reading

- **Ericsson, 5G, available at:** *https://www.ericsson.com/en/5g*
- **Ericsson, 5G voice, available at:** *https://www.ericsson.com/en/5g-voice*
- **Ericsson, 5G RAN, available at:** *https://www.ericsson.com/en/ran*
- **Ericsson, 5G Core, available at:** *https://www.ericsson.com/en/core-network/5g-core*
- **Ericsson, Network slicing, available at:** *https://www.ericsson.com/en/network-slicing*

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NETWORK COMPUTE FABRIC ✱

The network compute fabric

– ADVANCING DIGITAL TRANSFORMATION WITH EVER-PRESENT SERVICE CONTINUITY

As successive mobile network generations have made connectivity faster and more reliable, cloudification and related virtualization of the underlying infrastructure have revolutionized the IT domain, making affordable cloud services available for widespread use. In 6G, these two major trends come together to create the network compute fabric, the innovation platform of the future.

AZIMEH SEFIDCON,
WOLFGANG JOHN,
MILJENKO OPSENICA,
BJÖRN SKUBIC

In order to fully support emerging use cases such as the Internet of Senses [1], cyber-physical systems and connected intelligent machines [2], the foundation of future network platforms must be built on tight integration between reliable, deterministic connectivity and scalable, affordable and efficient processing capabilities. The result of this tight integration is what we call the network compute fabric.

■ The network compute fabric is one of the four key components of our vision of 6G as an innovation platform [3]. The other three are limitless connectivity, cognitive networks, and trustworthy systems, as shown in *Figure 1*.

The network compute fabric is a fusion of connectivity and computing capabilities, acting as one unified entity, to satisfy both compute and connectivity requirements. Currently, for instance, service anchoring, routing and channel termination are governed by the mobile radio network, while placement and allocation of application tasks are governed by evolving operations support systems and local operating systems. By tightly integrating networking and compute elements, functions like radio-channel handling would operate in concordance with the allocation of related application tasks, opening up for fine-grained optimization possibilities.

Components of the network compute fabric

The left side of *Figure 2* illustrates the four key components of a global, unified network compute fabric: unified ecosystems, unified application management, unified execution environment and unified exposure of network and compute capabilities.

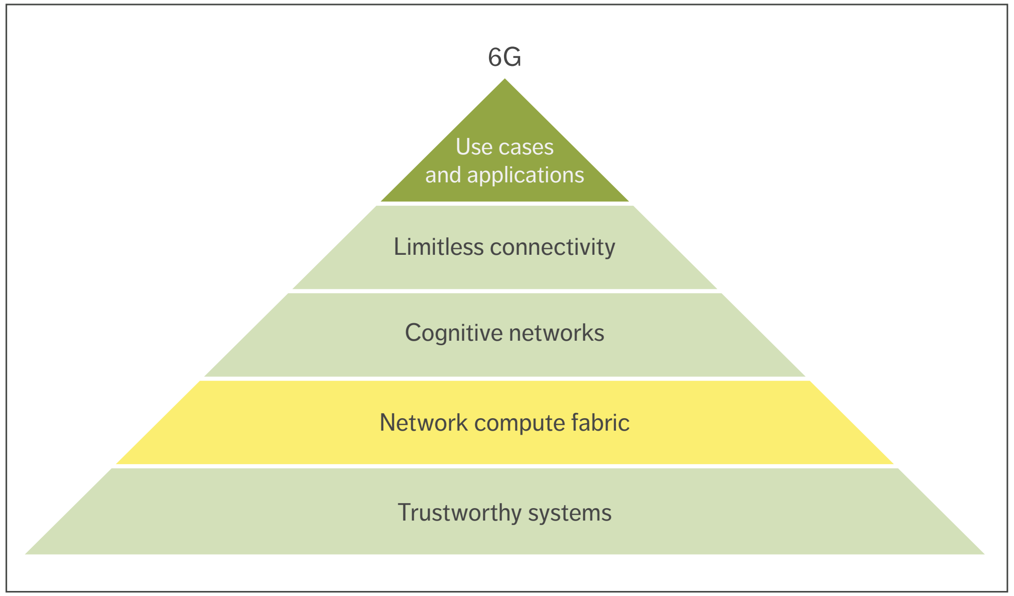

Figure 1 6G as an innovation platform for emerging use cases and applications

Unified ecosystems

To enable a wider range of more demanding use cases, the network compute fabric needs to facilitate the transformation of ecosystem engagements. This will be realized through the collaboration of a broad set of actors, driven by the need to mutualize the cost of infrastructure, offer services at larger scale and enable full roaming between providers. Network and cloud providers, application developers, service providers, and device and equipment vendors all have a role to play. Network operators have an opportunity to utilize their distributed network infrastructure as an innovation platform for richer service offerings, combining connectivity with compute in both wide-area and on-premises edge-deployment scenarios [4].

Several ecosystem models may evolve and have already been described in the context of edge computing [5]. There are four basic models for network compute fabric services: standalone, facilitation, aggregation and federation. The standalone model is where the network compute fabric services are offered by one actor in a standalone manner. The facilitation model is where one provider acts to unify fragmented offerings from other providers. The aggregation model is where one provider is active from a business perspective in building and reselling bundles of services from

Terms and abbreviations

CSP – Communication Service Provider | **E2E** – End-to-End

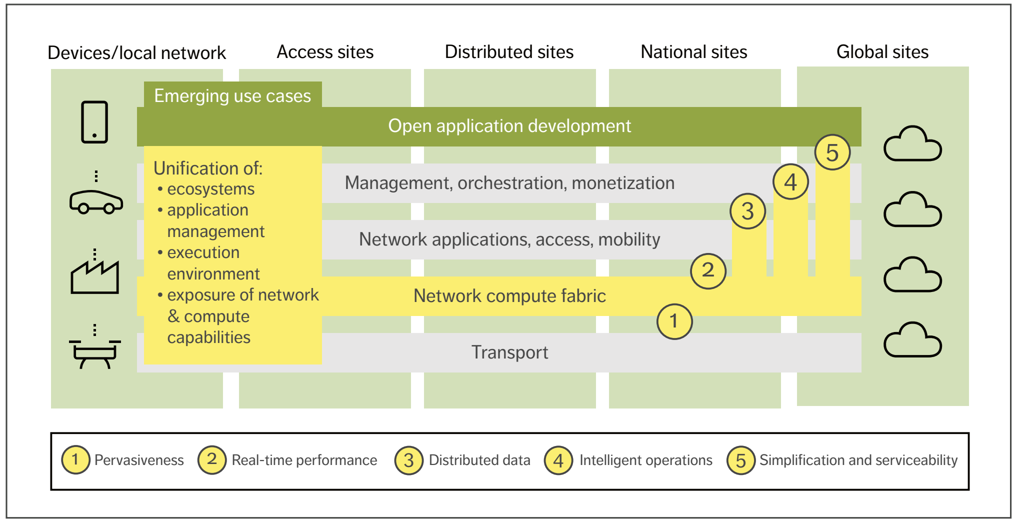

Figure 2 The four key components of the network compute fabric (on the left) enable a new set of features for both network and use-case applications (1-5 on the right)

multiple providers. The federation model is when the providers collaborate within a framework to utilize each other's offerings.

Ecosystem partnerships involve dynamic business agreements that need to be reached regarding the usage of resources between multiple partners. They also involve technical challenges of integrating services from the different actors. Ecosystem partnering can be facilitated by standards that ensure interoperability or by technologies that automate the handling of partner relationships. The right level of harmonization in the ecosystem is key to support scalability as well as innovation.

Unified application management

The network compute fabric will be a highly distributed platform that enables the execution of applications across multiple administrative domains. This innovation platform will require intelligent and data-driven operations to seamlessly span devices and network and cloud domains. An integrated DevOps toolchain will enable the appliance of similar development and operational methodologies across those domains. By separating orchestration from application functionality, the network compute fabric will allow tailored optimization for different network domains without the need for application code change.

Applications are usually composed of one or more services that may have different deployment and performance requirements. With interoperable abstractions and programming models, services can be easily deployed across different domains. With common operation models and tools, service quality can be assured autonomously per domain, considering joint network and compute resources and domain-specific performance requirements. Applications can also be collocated with other applications with dependent performance requirements. Common optimization loops will then enable operational synergies across such applications – that is, data-driven shared knowledge can proactively optimize dependent applications beyond individual optimization models.

Generalized operational pipelines and tools with programmable infrastructure will also enable the offloading of the common orchestration functionality from applications to the fabric. This will enable simpler application programming models with a stronger focus on the core application functionality.

With common data pipelines and data governance models including data protection, data flows can be combined across resource, functional and administrative domains for multi-layer data classifications that can bring aggregated and more accurate optimizations beyond single domain maturity.

Unified execution environment

The unified execution environment will act as an operating system, providing fundamental functionalities and services on top of distributed and heterogeneous network, compute and storage assets [6]. Evolving the ideas of serverless computing, it will facilitate the development and deployment of distributed applications on top of this infrastructure. The application has access to a compute service that always appears local, despite dynamic network changes or user/data mobility events. The unified execution environment will simplify the development of distributed applications by offering several capabilities.

Seamless task mobility will be enabled by evolved container formats based on portable bytecode such as WebAssembly and associated system interfaces. This will allow the platform to dynamically schedule workloads on nodes regardless of varying hardware and system software setups. As a result, several optimizations can be performed with limited overhead, such as moving computations close to a data source or consumer. This can be useful for a variety of reasons, including having computational tasks follow mobile users, offloading workloads from the device to preserve device energy and moving computations for an optimized cost/performance trade-off.

A distributed data infrastructure with a range of built-in consistency mechanisms will allow developers to choose from the right-sized trade-off between consistency levels and their associated resource costs. This will offer application tasks seamless access to data stored in multiple locations, as if from a single local access. As the application components will be distributed between several locations in a pipeline fashion, the platform would be in full control of both the compute and data locations, as well as traffic management and prioritization. As a result, data access can be optimized along the computation pipeline even considering network characteristics and usage profiles, while obliging legal or operational data policies.

Unified exposure of network and compute capabilities

The computational environment in the network compute fabric will be heterogeneous, which will increase with emerging computational hardware complementing traditional network equipment. One example is hardware acceleration technologies that provide compute and storage capabilities specialized for certain workload tasks, such as graphics processing units, field-programmable gate arrays and storage-class persistent memories.

By adapting tools and languages like OneAPI [7], developers will be able to write a single-source implementation of an algorithm that will be able to run on different hardware choices. Selection of actual processing or memory/storage instance will be based on availability and performance requirements, making it easy for developers to create performant applications that can run anywhere.

The network compute fabric also opens up for extensive in-network computation. Modern transport networking equipment will no longer be limited to packet transport, but also able to provide programmable compute capabilities, opening up for new divisions of application functionality where some tasks, such as QoS, scheduling, policing,

> THE RIGHT LEVEL OF HARMONIZATION IN THE ECOSYSTEM IS KEY TO SUPPORT SCALABILITY AS WELL AS INNOVATION

encoding and recoding can be performed directly in the data plane.

The network compute fabric will enable efficiency gains through transparent shortcuts inside the infrastructure between application components and the network and compute platform. Once applications are collocated with network functions on the same host, rack or cluster, parts of the network and operating system stacks can be bypassed. Depending on the situation, different technologies can be used, such as modern interconnect technologies and protocols in combination with fast replication services like Derecho [8].

Features of the network compute fabric

The unification of ecosystems, application management, execution environments and exposure of network and compute capabilities will lead to the creation of a network compute fabric that can deliver the five advanced features shown on the right side of Figure 2: pervasiveness, real-time performance, distributed data, intelligent operations, and simplification and serviceability.

Pervasiveness

The network compute fabric will provide ubiquitous compute resources, fully integrated with the network to complement connectivity, spanning from central parts all the way out to devices. This will enable distributed applications that interact with physical reality to benefit from having their tasks close to data sources and data consumers. Examples of where this will be useful include radio beamforming, closed-loop control of mission-critical processes and the intelligent aggregation of large amounts of data.

The network compute fabric will ensure optimized application execution with real end-to-end (E2E) performance guarantees, spanning both the connectivity and compute domains. In this context, pervasive should be interpreted as "everywhere" [9]. Billions of networked smartphones and powerful connected devices in combination with cloud and edge resources form an immensely powerful, unified compute infrastructure that has the potential to enable completely new, unprecedented applications and services that could not be easily accomplished with traditional, siloed infrastructures.

> THE NETWORK COMPUTE FABRIC WILL ENSURE OPTIMIZED APPLICATION EXECUTION WITH REAL END-TO-END PERFORMANCE GUARANTEES

Real-time performance

Current and emerging use cases often require a combination of stringent real-time characteristics, such as low latency, high throughput, high reliability and scalability. E2E performance requirements demand that deterministic and reliable connectivity offerings are complemented with corresponding compute capabilities. Connectivity with an integrated real-time compute stack will enable the network platform to host and manage the critical tasks of real-time applications to guarantee true E2E performance.

Use cases such as robotics and ubiquitous augmented reality or virtual reality control imply strict latency bounds in combination with distribution and elasticity requirements. These are difficult to achieve with segmented control loops of individual platform components and application tasks. Additionally, virtualization technologies predominant in the cloud often add latency, decrease throughput and cause jitter in data flows. Such unpredictable characteristics affect the E2E performance of applications.

Edge computing involves bringing computing capabilities either closer to data sources or to application front ends. By building on this principle and taking it even further, utilization of shorter control loops will eliminate latency toward central services. However, data packets still pass through numerous layers of network and virtualization layers with fragmented control loops. A truly reliable real-time solution requires coordinated control of network and compute capabilities in a holistic way with full awareness of the application. Moreover, hypervisors and the network must operate in concert and be synchronized to avoid jitter and timing variations. This will require the network and compute components to be tightly related.

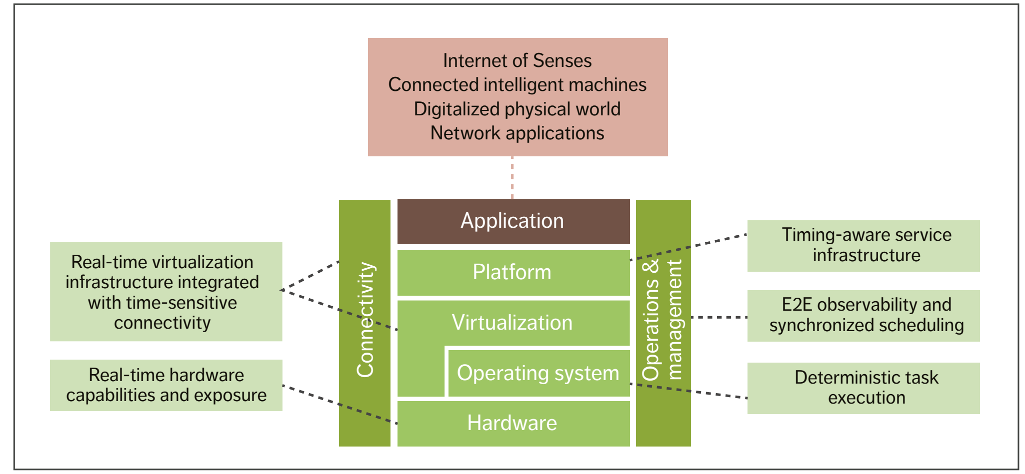

Figure 3 Real-time cloud stack supporting the network compute fabric performance characteristics

New network technologies such as Ultra-Reliable Low Latency Connectivity provide predictable wireless communication, while Time-Sensitive Networking for wired communication can bring network latency down to a few milliseconds. A compute infrastructure building on time-sensitive networks will integrate real-time properties on all layers of the stack, as shown in *Figure 3*. Data-plane services such as messaging and storage will provide guarantees on latency bounds, and services on the control and management planes have to support time-sensitive operations accordingly. Key components of such fabric are predictable networking and compute technologies, coupled with advanced resource management and scheduling features, realized at the operating system and hypervisor levels.

Distributed data

Unified data management together with computing across the network is the key to reducing data latency and transfer volumes while not impacting the quality of data-driven applications including artificial intelligence. The network compute fabric offers capabilities to utilize the available compute, storage and network resources more efficiently.

Allowing data to be processed closer to the data source significantly improves round-trip times and time to action/response for mission-critical use cases. Raw data does not have to be transmitted across the network but can be processed locally, and thereby be transformed to models or insights. As a consequence, only necessary real-time business insights, equipment maintenance predictions or other actionable data would be shared across network locations, as input to further federated learning.

Furthermore, the existence of a distributed data infrastructure within the unified execution environment makes it possible to utilize data shortcuts when application tasks access state or other types of data. These shortcuts can be realized with the help of modern cluster interconnect technologies that bypass the kernel and parts of the networking stack, such as RDMA (Remote Direct

> AUTOMATED MANAGEMENT LOOPS WILL MAKE THE FABRIC THE FOUNDATION OF NEW AUTONOMOUS APPLICATIONS

Memory Access), PCIe NTB (Peripheral Component Interconnect express Non-Transparent Bridging) and so on.

A distributed data infrastructure also makes it possible to offer advanced data services such as local caching mechanisms and data-centric near-memory and near-storage computing technologies, which can be very convenient for both developers and operations.

Intelligent operations

Manual and reactive operation practices often lead to suboptimal efficiency and performance. In the automated systems that are in use today, business intent is translated centrally to predefined workflows and then executed by individual domains. To achieve optimal efficiency and performance, however, it is essential to manage orchestration and operations across the entire ecosystem. This can be achieved by enhancing automated control and management processes, and applying data-driven learning and cognitive functions as part of unified cross-domain optimization models. With the network compute fabric, new operational models that can proactively evolve with the system will be able to manage the dynamics of network payloads and application workloads in real time.

The network compute fabric will utilize cognitive capabilities and data-driven automation to manage synergies across network, compute and storage domains. Distributed intelligence based on telemetry data from the network and compute infrastructure as well as application performance indicators will realize short feedback cycles for instant run-time optimizations of application deployment and infrastructure configurations.

Automated and self-maintained management loops will make the fabric the foundation for new autonomous applications. Examples of automation steps include:

- Predictive deployment strategies for applications and platform services
- Predictive resource availability and collocation for different domains
- Intelligent sharing and scheduling of rare resources such as hardware accelerators
- Qualitative trade-off analysis of conflicting business and operational intents
- Autonomous application deployment and life-cycle management
- Predictive fault management and platform resiliency.

Simplification and serviceability

The placement decisions for the distributed application components will be based on service coverage, E2E service requirements and infrastructure capabilities. The placement will also be adapted dynamically based on conditions of resource availability.

Such dynamic distribution of application components places a new set of requirements on the serviceability of applications. For instance, traffic flows between individual application components have to be managed automatically in order to maintain ongoing sessions. DevOps pipelines across ecosystem partners need to be an integral part of the fabric. This includes developer-friendly tracing and debugging capabilities for vastly distributed applications. These types of features and services need to be built into the fabric and exposed through convenient application programming interfaces to the application operators and developers.

The network compute fabric will offer possibilities to specify high-level business intents for services or applications. For instance, simple low-code/no-code interfaces will allow application providers to declare requirements associated with the deployment of the distributed application. Across the service graph of an application, there may be performance requirements of varying stringency that make it possible for the fabric to automatically strike a balance between application performance requirements and operational costs.

Business opportunities of the network compute fabric

While the monetizable differentiators for each player will vary significantly, the network compute fabric will enable communication service providers (CSPs), developers and many other types of players to offer new services in a wide variety of contexts, including the Internet of Senses and connected intelligent machines.

Depending on the interplay of harmonized telecom and IT domains, these differentiators could be monetized based on the services the mobile network is expected to provide. The majority of 6G use cases will generate massive amounts of data, which will enable novel insights using the intelligent algorithms embedded in the network compute fabric.

The 6G world will require a shift in how CSPs do business in terms of exposing services to new types of customers. These services need to be bundled or aggregated from multiple domains into relevant offerings. The offering may range from pure connectivity services, to integrated connectivity and compute, to other value-added services. These types of services may be exposed directly to the customers or delivered by ecosystem partners that aggregate services for customers.

The network compute fabric will enable new business models by integrating the services of different ecosystem actors and providing new technologies for as-a-service models. Automated contract negotiation and fulfillment to support sales, delivery and charging operations can reduce business friction within various ecosystem constellations. Much of the interaction between the players will happen in software. New technologies based on distributed ledgers and smart contracts can enable new brokerless partnership models between providers.

Application-supporting services can play an important role in the ecosystem by simplifying application development for application providers. Further development of technologies and synergies across ecosystems will allow for additional business model innovation.

Conclusion

Our vision of the network compute fabric is based on expected 6G use cases such as the Internet of Senses and cyber-physical systems that will require a new set of capabilities beyond connectivity. These capabilities will not only serve demanding use cases but will also transform the network into an innovation platform characterized by its cognitive behavior, trustworthiness and adaptability.

The principal components of the network compute fabric facilitate unification across ecosystems, application management, execution environment, and exposure of network and compute capabilities, regardless of the heterogeneity of the infrastructure. The key features of the network compute fabric are pervasiveness, real-time performance, distributed data, intelligent operations, and simplification and serviceability.

The network compute fabric will be a pervasive, globally interconnected compute and storage platform that facilitates optimized handling of application components while giving the impression of locality. Real-time infrastructure and services together with unified data access are core elements of the network compute fabric, supporting distributed intelligence as well as simplification of deployment across heterogeneous infrastructure and administrative domains.

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Further reading

- **Ericsson website, Network compute fabric, available at:** *https://www.ericsson.com/en/future-technologies/network-compute-fabric*
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XR and 5G: Extended reality at scale with time-critical communication

The value of 5G extends far beyond the enhanced mobile broadband that more than 1 billion people already have access to through upwards of 150 communication service providers around the world [1]. The time-critical communication capabilities in 5G networks will enable major breakthroughs in a wide range of application areas, including extended reality (XR).

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In contrast to enhanced mobile broadband (MBB), the majority of the emerging 5G value-adding applications are time critical in nature with demanding requirements on reliable low latency. Time-critical use cases can be broken down into four categories: real-time media, remote control, industrial control and mobility automation [1, 2].

■ The real-time media category includes innovative extended reality (XR) applications that are of growing interest for consumers, enterprises and public institutions alike. The emergence of time-critical communications over 5G networks makes it possible to offload parts of the XR processing and functionality to the edge cloud and enhance the user experience with lightweight and cost-efficient head-mounted displays (HMDs).

XR use cases

XR is an umbrella term that covers immersive technologies ranging from virtual reality (VR) to mixed reality (MR) and augmented reality (AR). In VR, users are totally immersed in a simulated digital environment or a digital replica of reality. MR includes all variants where virtual and real environments are mixed. AR is one such variant, where digital information is overlaid on images of reality viewed through a device. The level of augmentation can vary from a simple information display to the addition of virtual objects and even complete augmentation of the real world. MR can also include variants where real objects are included in the virtual world.

XR is expected to improve productivity and convenience for consumers, enterprises and public institutions in a wide variety of application areas such as entertainment, training, education, remote support, remote control, communications and virtual meetings. It can be used in virtually all industry segments, including health care, real estate, shopping, transportation and manufacturing. VR is already used for gaming both at home and at dedicated venues, for virtual tours in the context of real estate, for education and training purposes and for remote participation at live events such as concerts and sports.

While VR holds great promise, AR and MR use cases have even greater transformational potential. In VR, the headsets cut users off from their physical surroundings and restrict mobility [3]. With AR, users are present in reality and free to move even when using HMDs. Many smartphone users have already experienced basic forms of AR, through games like Pokémon Go and apps that enable shoppers to visualize new furniture in their homes before making a purchase. AR technology becomes much more powerful, though, when it is used with HMDs. By freeing up the user's hands, AR HMDs transform the user interaction. The ability to have information overlaid on the real world while simultaneously having your hands free has been shown to increase worker efficiency dramatically [4].

> WHILE VR HOLDS GREAT PROMISE, AR AND MR USE CASES HAVE EVEN GREATER TRANSFORMATIONAL POTENTIAL

XR edge-processing architectures

VR HMDs are already available at scale on the market today, but they are rapidly evolving. Because VR applications are often processing-intensive, enabling high-end VR requires connecting the HMDs to high-end processing units – typically powerful PCs or gaming consoles. VR HMDs with local processing capabilities are also starting to become available, but these are relatively large and heavy and cannot provide the same experience as when off-device processing is used.

AR HMDs are also available on the market today, predominantly for enterprise use [5]. Mass-market adoption will require further progress on ease of use, attractive appearance and content availability [6]. Building fashionable, small form factor HMDs to meet the demands for XR is challenging due to limited processing power, storage, battery life and heat dissipation. We believe that the best way to address these challenges is by offloading parts of XR processing to the mobile network edge.

Terms and abbreviations

5GC – 5G Core | **AAS** – Advanced Antenna System | **AR** – Augmented Reality | **CG** – Configured Grant | **CoMP** – Coordinated Multi-Point | **DAPS** – Dual Active Protocol Stack | **DC** – Data Center | **DL** – Downlink | **E2E** – End-to-End | **L4S** – Low Latency, Low Loss, Scalable Throughput | **HMD** – Head-Mounted Display | **MBB** – Mobile Broadband | **MR** – Mixed Reality | **SPS** – Semi-Persistent Scheduling | **TRP** – Transmission Reception Point | **TTI** – Time Transmission Interval | **UE** – User Equipment | **UL** – Uplink | **VR** – Virtual Reality | **XR** – Extended Reality

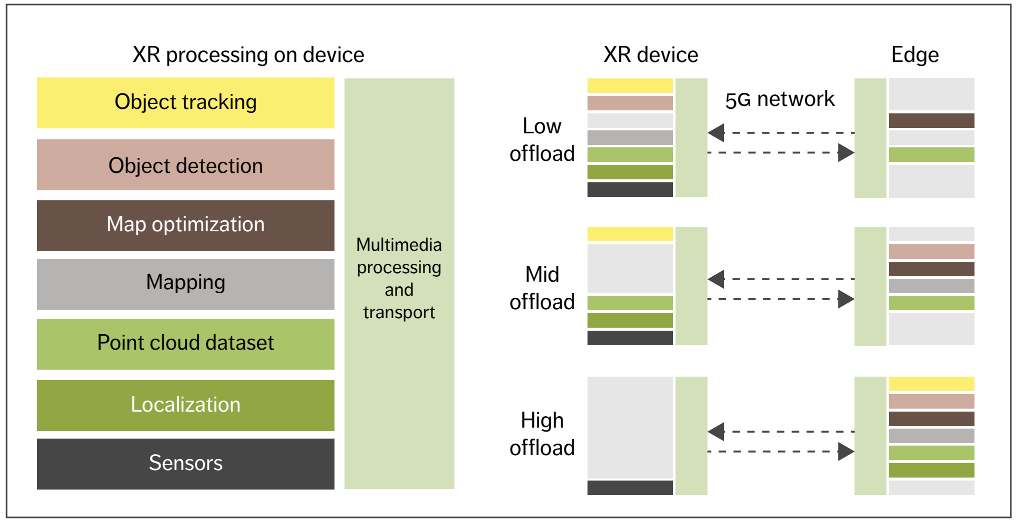

Figure 1 Split architecture options with 5G connectivity

Figure 1 shows the eight main types of XR functionality and how they can be split up between an XR device and the network edge. The major components in XR processing include SLAM (simultaneous localization, mapping and map optimization) [7] with point cloud datasets [8], hand gesture and pose estimation [9], object detection and tracking [10] and multimedia processing and transport. Examples of multimedia processing and transport are rendering, asynchronous time warp [11] and video, audio and sensor encoding.

Figure 1 also illustrates three architectures for splitting the XR processing: low offload, mid offload and high offload. Our internal studies indicate that low offload reduces device energy consumption by threefold while mid offload reduces it by fourfold. High offload reduces device energy consumption by more than sevenfold.

OUR INTERNAL STUDIES INDICATE THAT HIGH OFFLOAD REDUCES DEVICE ENERGY CONSUMPTION BY MORE THAN SEVENFOLD

In the low-offload architecture, almost all processing is done on the device. The point cloud dataset, spatial map generation and localization occur on the device. As portions of the point cloud and spatial map are built, the point cloud is compressed through multimedia processing and transmitted over the 5G network to the edge, where uploaded spatial map data is created and merged into existing global spatial map data. The object detection and tracking are performed on the device. The rendering can occur locally on the device or at the network edge.

In the mid-offload case, the localization and object tracking functions are performed on the device. The spatial map and point cloud datasets are generated in the network. Key image frames are compressed using a video codec and sent to the network edge to generate the spatial map and point cloud datasets, and to perform object detection. At the network edge, the point cloud datasets and spatial map are created and merged with the global point cloud datasets and spatial map. The overlay rendering occurs at the network edge. The edge-rendered video and audio content is encoded into video and audio streams and transmitted to the device along with the rendering mesh data.

In the high-offload case, only sensor data is sent over the uplink. Sensor data includes camera data. Many AR/VR devices have multiple cameras, including infrared and RGB (red, green and blue) ones. Sensor data also covers sensors such as LiDAR (light detection and ranging) and IMU (inertial measurement unit). The image data is encoded using video compression such as Motion Pictures Experts Group (MPEG) High Efficiency Video Coding (HEVC) or Versatile Video Coding (VVC).

THE XR CONNECTIVITY REQUIREMENTS DEPEND ON THE LEVEL OF SPLIT ARCHITECTURE AND THE TARGETED QoE

Several compression techniques have been proposed for IMU data, such as delta encoding, linear extrapolation, second- to fifth-order polynomial regression and spline extrapolation [12]. There are numerous techniques for compressing 3D point cloud generated by LiDAR sensors. ISO/MPEG currently has two tracks for point cloud compression standardization under development. These are Video-based Point Cloud Compression (V-PCC) and Geometry Based Point Cloud Compression (G-PCC). LiDAR is covered in the MPEG G-PCC track [13]. The edge-rendered video and audio content is encoded into video and audio streams and transmitted to the device along with the rendering mesh data.

XR traffic characteristics and connectivity requirements

XR traffic is characterized by a mixture of pose and video from/to the same XR device, varying video frame size over time and quasi-periodic packet arrival with application jitter after IP segmentation. Traffic arrival time to the RAN is periodic with non-negligible jitter due to application-processing-time uncertainty. Video frame sizes are an order of magnitude larger and, at the same time, not fixed over time compared with packets in voice or industrial control communication. The segmentation of each frame is expected, which implies that packets arrive in bursts that must be handled together to meet stringent bounded latency requirements.

XR connectivity requirements depend on the level of split architecture and the targeted QoE, leading to a wide range of bit rates and bounded latency requirements. *Figure 2* presents the 5G connectivity requirements for AR, VR and cloud gaming based on the developments in the ecosystem, including the 3GPP [14]. The requirements assume local processing techniques in the split architecture to

Use cases	DL bitrates (Mbps)	UL bitrates (Mbps)	One-way latency (ms)	Frame realibility (%)
Cloud gaming	8-30	~0.3	10-30	≥99
VR	30-100	< 2	5-20	≥99
AR	2-60	2-20	5-50	≥99

Figure 2 Use-case requirements for 5G networks

EXTENDED REALITY AND 5G ✱

mitigate consumer latency requirements [15]. Note that latency and reliability requirements are on a video frame (or file) level excluding application error and delay.

For downlink (DL) video traffic, VR typically needs higher bit rates than cloud gaming to support retinal resolution when using HMD and lower compression efficiency due to low-latency encoding. Some AR applications for conversational services can have DL video traffic as VR. However, they potentially have lower resolution to render a video on only part of display, leading to a lower bit rate than VR DL video. In addition, they can have uplink (UL) video streaming for the object detection and tracking, but the bit rate requirements can be lower compared with the DL video. All cloud gaming, AR and VR include pose traffic in the UL, which has much lower bit rates than video traffic, but AR and VR will have higher bit rate requirements than cloud gaming to convey more pose information, such as six degrees of freedom.

AR and VR require more stringent end-to-end (E2E) bounded latencies than cloud gaming since a human is more sensitive to the discrepancy in 3D virtual environments. For instance, it is widely accepted that rendering motion to photon latency greater than 20ms starts to cause nausea when a human wears a VR HMD. Processing techniques such as asynchronous time warp [11] relax the latency requirement to some extent, making VR feasible over 5G networks. For AR, object detection can be performed at the network edge, and object tracking can be done on the device as shown in the low- and mid-offload cases in Figure 1. By performing object detection in the network and tracking on the device, the bounded latency requirement for a 5G network can be relaxed up to 50ms.

Network architecture for time-critical communications

Time-critical communications is an emerging 5G concept for enabling services with reliable low latency requirements such as XR [2]. The aim is to secure data delivery within specific latency bounds (X ms) with the desired reliability level (Y percent). Depending on the user requirements, X ranges from tens of milliseconds to 1 millisecond latency and Y ranges from 99 percent to 99.999 percent reliability. To ensure bounded latencies, the system may have to compromise on capacity, throughput, energy efficiency or coverage.

The 5G RAN, the 5G Core (5GC) and the transport network together with the device contribute to the E2E reliability and latency. E2E latency is the sum of individual latency contributions from every component. E2E reliability cannot be better than the reliability of the weakest link.

Edge deployment of 5GC and applications is key to reducing the transport latency between the application and the RAN. If an application is hosted in a central national data center (DC), the transport network round-trip latency can be in the order of 10-40ms, depending on the distance to the DC and how well the transport network is built out. The transport latency can be reduced to 5-20ms by moving applications to a regional DC or even to 1-5ms for edge sites. For local network deployments with networking functions and applications hosted on-premises, transport latencies become negligible [2].

Achievable RAN latency/reliability performance depends on general deployment factors (such as frequency band, bandwidth, inter-site distances, numerology, duplexing schemes and TDD configuration), RAN and user equipment (UE) capabilities (in terms of hardware and software features) and traffic characteristics (such as data rate and packet size).

5G toolbox to achieve time-critical communications

To ensure that XR applications work well over 5G networks, it is important to separate the XR traffic from best-effort MBB traffic using the 5G QoS framework with optimized QoS flows, as illustrated at the top of *Figure 3*. This enables optimized treatment of XR throughout the mobile network and specifically in the RAN to mitigate the different sources of delay.

The provision of bounded latency for XR and

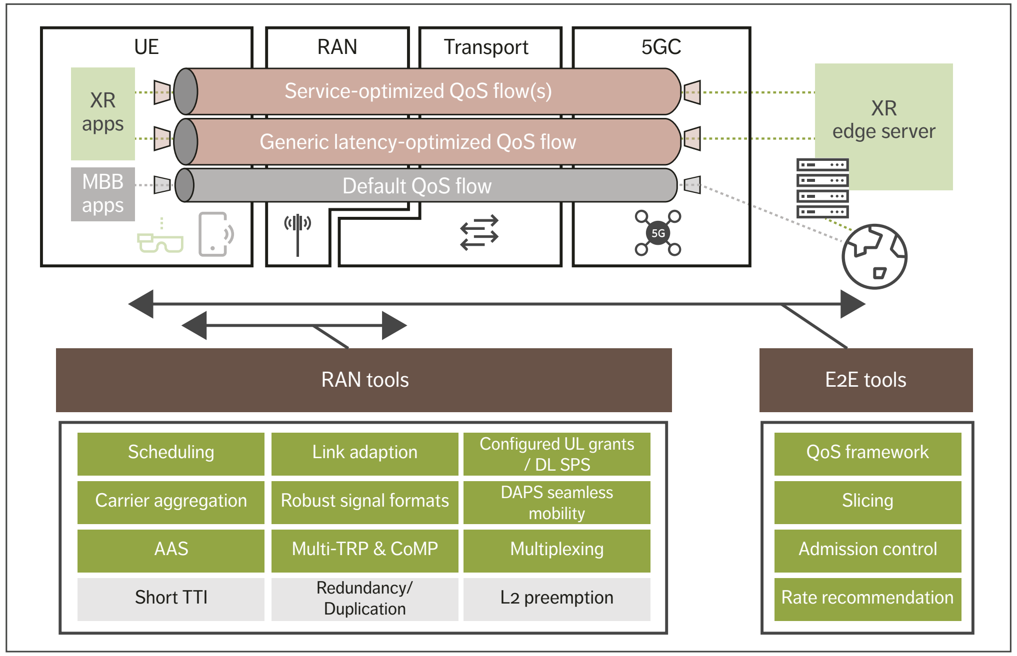

Figure 3 5G toolbox to realize time-critical communications – the tools that are important for XR are marked in green

other services requires the prevention of delays and interruptions. The bottom half of Figure 3 presents the comprehensive 5G toolbox for realizing time-critical communications that makes it possible to overcome the following five main causes of delays and interruptions:

1. Congestion
2. Dynamic radio environment
3. Standards/protocols
4. Mobility
5. Device power saving.

Congestion

There are several ways to mitigate the congestion-related delays that can occur when end hosts transmit at a higher bitrate than the network can sustain. Fast application rate adaptation is key to avoid congestion for rate-adaptive traffic such as XR and cloud gaming. Low Latency, Low Loss, Scalable throughput (L4S) is an existing method that provides fast indication of congestion from networks that can be applied in the RAN, enabling RAN rate recommendation and forming the basis for a common framework for rate adaptation over 5G [16]. Network slicing and radio resource partitioning together with admission control and latency-optimized scheduling are important tools for reserving network resources for time-critical services and avoiding congestion-related delays to provide a minimum guaranteed bit rate, for example. Network slicing is also beneficial for protecting MBB and other services from resource-hungry, time-critical services.

Dynamic radio environment

Advanced scheduling together with robust link adaptation and signal transmission formats are important tools to combat delays related to the

> THE 5G QoS FRAMEWORK MAKES IT POSSIBLE TO ESTABLISH QoS FLOWS THAT PROVIDE OPTIMIZED NETWORK TREATMENT FOR SPECIFIC FLOWS

dynamic radio environment such as fading/blocking and interference. High-rate XR traffic places new demands on these tools to deliver very high spectral efficiency, while ensuring reliable and timely video frame delivery. Advanced antenna systems (AASs) have tremendous potential to improve the link budget, reduce interference, and increase spatial multiplexing, ultimately leading to more radio system capacity for XR.

Standards/protocols

Features that minimize delays associated with standards/protocols include tools like prescheduling, UL configured grants (CGs) and DL semi-persistent scheduling (SPS). As an example, a combination of periodic CGs and dynamic grants can be used to reduce latency for UL AR traffic consisting of periodic frames of varying size. XR traffic arrival time characteristics call for further optimizations of UL CGs and DL SPS to avoid unnecessary waiting times. Protocol enhancements throughout the different protocol layers, from the SDAP (Service Data Adaptation Protocol) to the physical layer, including the PDCP (Packet Data Convergence Protocol), the RLC (Radio Link Control) and the MAC (Medium Access Control) protocols, can be important for improving the capacity of XR applications as well. For example, control signaling efficiency could be improved to provide grants for multiple radio resource allocations needed for a large XR video frame.

Mobility

Time-critical services place much more stringent requirements on mobility performance than MBB services. For XR services, mobility interruptions need to be well below the inter-frame arrival time (which is typically between 15-50ms) to pass unnoticed. This will require smarter and faster network algorithms as well as stricter processing requirements at the device. 5G New Radio provides a few options for supporting seamless and more robust mobility, including multiple transmission reception points (multi-TRPs), dual active protocol stack (DAPS) handover and conditional handover.

Device power saving

Device power saving is important to support low-power XR devices, and XR traffic arrival time characteristics also call for further optimizations of discontinuous reception.

Traffic awareness in the RAN

One interesting area of future standardization work in the 3GPP is to further optimize 5G performance and capacity by improving XR traffic awareness in the RAN, especially for very short-term traffic variation that may be difficult for an application layer to handle. If, for example, the RAN had the ability to know which IP packets are associated with the same application frame, it could potentially use that knowledge to optimize radio resource allocation, scheduling, link adaptation, packet discarding and other features to increase capacity.

5G QoS approaches for extended reality

The 5G QoS framework makes it possible to establish QoS flows that provide optimized network treatment for specific traffic flows, in addition to the default QoS flow used for MBB. Such additional QoS flows can be established either using 5GC QoS-exposure application programming interfaces to communicate service requirements, or by traffic detection together with pre-provisioned service requirements, such as relying on standardized 5G QoS identifier characteristics. Two complementary 5G QoS approaches can be implemented for XR: a generic latency-optimized QoS flow and service-optimized QoS flows.

RAN deployment	Frequency allocation	Min. bit rate	Max. latency	Frame reliability
Wide area	Mid-band FDD 2x20MHz@2GHz	DL: 8-30Mbps UL: 2-10Mbps	DL: 10-30ms UL: 10-30ms	99%
	Mid-band TDD 100MHz@3.5GHz			
Indoor	Mid-band TDD 100MHz@3.5GHz	DL: 30-60Mbps UL: 10-20Mbps	DL: 10-30ms UL: 10-30ms	99%
	mmWave 800MHz@30GHz		DL: 5-10ms UL: 5-10ms	

Figure 4 Simulation assumptions for 5G RAN

Generic latency-optimized QoS flow

A generic latency-optimized QoS flow that can be implemented in any network is an important tool to enable a large ecosystem of high-rate, rate-adaptive, time-critical applications and services including XR to emerge and hopefully flourish in the same way as MBB and smartphones have done [16]. Implementing such a generic QoS flow enables networks to provide L4S-based early RAN congestion detection and fast rate recommendation to applications as well as packet treatment optimized for low latency and jitter rather than throughput. By not relying on knowledge of specific service requirements it avoids a tight coupling between application and network.

Service-optimized QoS flows

Service-optimized QoS flows can be established for specific XR services with known service requirements in terms of packet delay budgets, packet error rates, minimum guaranteed bit rates and so on, in cases where there is a need to increase coverage or capacity while still providing a good minimum QoE, for example. This requires a tighter coupling between application and network, which adds complexity to the ecosystem but provides QoE beyond what the generic latency-optimized QoS flow can enable and may be required for the most demanding XR applications.

Deployment strategy

To develop insights regarding the 5G network deployment strategy for various XR applications, we have carried out simulation studies for a wide-area deployment and an indoor enterprise deployment. For the wide-area scenario, we evaluated capacity for low- to mid-range XR applications in terms of requirements. For the indoor deployment, we considered high-end applications as well. The simulation assumptions are summarized in *Figure 4*. The 99 percent frame reliability implies less than 0.1 percent packet error rate if each frame is segmented into more than 10 IP packets.

The wide-area scenario is based on a macro deployment in central London with an inter-site distance of approximately 450m. For the mid-band, wide-area deployments, we include an AAS with 32 elements and 16 elements per polarization for 3.5GHz and 2GHz, respectively. The indoor deployment also assumes eight elements and 32 elements per polarization of an AAS for 3.5GHz and 30GHz, respectively. Devices with four receiver

EXTENDED REALITY AND 5G ✱

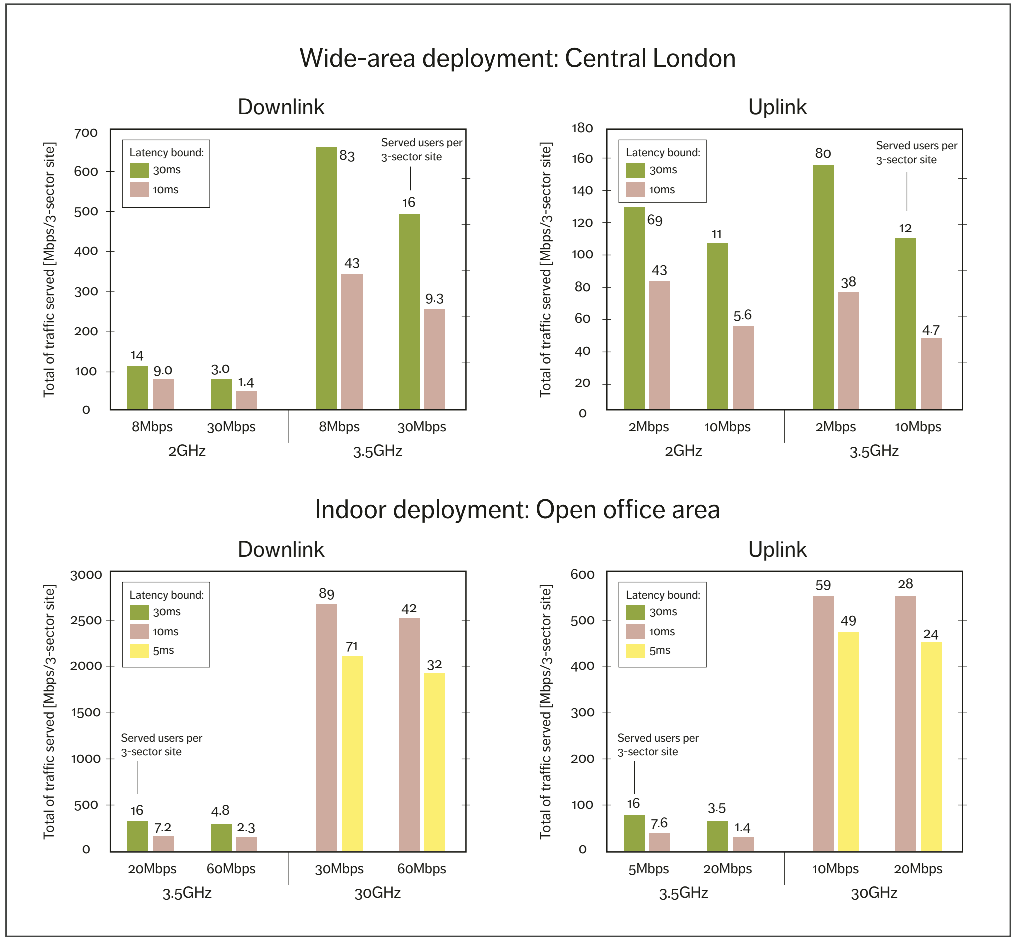

Figure 5 5G RAN-served traffic for various requirements

branches were used in the evaluation. The indoor deployment assumes a 120m by 50m open office area, where four pico sites with three sectors each are placed on the ceiling. For the TDD bands, we have assumed 4:1 DL and UL configuration.

Figure 5 shows the average served traffic per cell (with different combinations of the minimum bit rate and maximum latency requirements per user) for the wide-area and the indoor deployment scenarios assuming 90 percent and 95 percent coverage availability, respectively. From the cell throughput and the minimum bit rate per user, it is possible to derive the maximum number of users served simultaneously per cell, which is shown next to each bar in Figure 5. When the cell capacity is underutilized, the applications can increase the bit rate for superior QoE by rate adaptation.

We can observe that the cell capacity in the traffic served decreases by increasing the minimum user bit rate requirements or by reducing the maximum latency target. Increasing the user bit rate requirement at a given latency budget creates more interference and resource utilization, leading to a smaller amount of total traffic being served. Similarly, when reducing the latency requirement for a given bit rate requirement, a more stringent latency target implies that the resources are available on a shorter time scale, which leads to fewer users being served.

These results show that communication service providers can start to address cloud gaming and low-end AR use cases (for consumers and enterprises) in a wide area with the ongoing 5G network rollout utilizing mid bands, and gradually evolve capabilities and densify deployments to address greater coverage and more demanding requirements in terms of throughput and bounded latency. For users with coverage needs in a small geographic area such as a theme park, industry campus or office, there is an opportunity to support high-end XR with indoor deployments and address more stringent latency and higher bit rate requirements with a local breakout of the core network.

Conclusion

Although extended reality (XR) has the potential to be transformational for both business and society, widespread adoption has previously been hindered by issues such as heat generation and the limited processing power, storage and battery life of small form factor head-mounted devices. The time-critical communication capabilities in 5G make it possible to overcome these challenges by offloading XR processing to the mobile network edge. By capitalizing on their ongoing 5G rollouts, mobile network operators are in an excellent position to enable the realization of XR on a large scale.

Further reading

- **Switch on a better 5G network, available at:** *https://www.ericsson.com/en/5g/5g-networks*
- **5G by Ericsson, available at:** *https://www.ericsson.com/en/5g*
- **How 5G and Edge Computing can enhance virtual reality, available at:** *https://www.ericsson.com/en/blog/2020/4/how-5g-and-edge-computing-can-enhance-virtual-reality*
- **A technical overview of time-critical communication with 5G NR, available at:** *https://www.ericsson.com/en/blog/2021/2/time-critical-communication--5g-nr*

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